

# Frequent Stress Tests\*

Deepal Basak   Mayur Choudhary   Zhen Zhou<sup>†</sup>

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## Abstract

Bank supervisors employ stress tests and disclose valuable information about banks to the stakeholders. Such tests are also conducted frequently. A credible stress test design must incorporate the stakeholders' strategic responses to test results and ensure that a bank that passes a stress test does not subsequently fail, even under the most adverse equilibrium. To study credible stress tests, we analyze bank run as a canonical regime change game under incomplete information and asynchronous moves. We find that a higher testing frequency reduces the need for stricter test stringency to maintain credibility. We then characterize the optimal credible stress test policy and demonstrate its dependence on model primitives.

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**JEL Classification:** D83, E58, G01, G28

## 1 Introduction

Silicon Valley Bank (SVB) was not due for its next supervisory stress test until 2024, and it failed much before that in March 2023. This exposed critical weaknesses in the U.S. bank supervision framework. The Federal Reserve's internal review underscored these deficiencies, noting that

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<sup>†</sup>Basak: Kelley School of Business, Indiana University, Email: [dbasak@iu.edu](mailto:dbasak@iu.edu); Choudhary: Kelley School of Business, Indiana University, Email: [mchoudh@iu.edu](mailto:mchoudh@iu.edu); Zhou: PBC School of Finance, Tsinghua University, Email: [zhouzh@pbcfsf.tsinghua.edu.cn](mailto:zhouzh@pbcfsf.tsinghua.edu.cn)

*“the supervision of SVB did not work with sufficient force and urgency.”*<sup>1</sup> This raises a natural question – when can we say that a stress test policy is sufficiently forceful (stringent) and urgent (frequent) such that—even in the most adverse scenario—it deters stakeholders from running on a bank that passes a test? We call such stress test policies *credible*. We characterize the set of credible stress test policies in a canonical regime change game, with incomplete information and asynchronous stakeholder actions, and derive the optimal credible stress test policy.

A more stringent test is harder to pass, and accordingly, passing a more stringent test sends a stronger signal. It is well understood that in a static bank run game, a single stress test is credible if and only if it is sufficiently stringent, and so, the regulator optimally chooses the least stringent test out of all credible tests.<sup>2</sup> If all of the stakeholders do not move at the same time, as is often the case in reality, then the regulator might conduct multiple tests. Conducting these stress tests is costly, and in some jurisdictions, tests are conducted more frequently than others.<sup>3</sup> However, the extant literature does not provide much clarity regarding the benefit of conducting more tests. While passing two tests rather than one test is a stronger signal, it is unclear whether the presence of the second test reassures the stakeholders who move before the second test, or instead heightens their concerns that a bank, despite passing the first test may fail the next one. In the latter case, they might attack more aggressively, causing the bank to fail the next test. Our analysis shows that *by increasing the frequency of stress tests, regulators can reduce the stringency of each individual test while still maintaining credibility, thereby allowing more banks to avoid runs*. This establishes the precise role of frequency as a policy lever in the design of stress tests.

In practice, supervisory stress tests evaluate whether banks can absorb losses under adverse but plausible macro-financial scenarios. Test results are typically binary—pass or fail—and publicly disclosed. The design of a stress test must incorporate stakeholders’ endogenous actions, which are shaped by their beliefs about the bank’s fundamentals and their expectations of others’ behavior—both of which are influenced by the stress test outcome. From the regulator’s standpoint, the credibility of test results is paramount: a passing grade must reliably signal that the bank will remain solvent even under stress. A regulator can design a stress test policy that the bank can pass easily; however, this may not be credible. If the stakeholders are not convinced by such a weak test, they may run, causing a bank failure. A non-credible stress test policy *“could shake confidence in the entire stress testing regime, calling into question not just*

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<sup>1</sup>“Review of the Federal Reserve’s Supervision and Regulation of Silicon Valley Bank,” <https://www.federalreserve.gov/publications/files/svb-review-20230428.pdf>.

<sup>2</sup>See, for instance, Goldstein and Huang (2016).

<sup>3</sup>For instance, the Federal Reserve conducts annual Dodd Frank Act Stress Test (DFAST), the European Banking Authority (EBA) relies on biennial system-wide tests, and the Bank of England has varied both the severity and scope over time. See Hirtle and Lehnert (2015) and Goldstein and Leitner (2018) for a discussion of design differences across economies.

*the resiliency of other participating banks, but also of the competence of the supervisory authorities.”*<sup>4</sup> For instance, the failure of Bankia and other Spanish *cajas* in 2012 after passing prior European Banking Authority (EBA) stress tests damaged the reputation of European regulators to such an extent that Oliver Wyman and Roland Berger were brought in to conduct independent tests of the Spanish banking system.

Formally, we model a stress test as a dynamic information disclosure policy characterized by two features: test frequency ( $N$ ) and test stringency ( $k$ ) of each pass/fail test, to which the regulator commits in advance. A distress event lasts for a fixed duration, during which a unit mass of agents sequentially decides whether or not to run on the bank (i.e., attack the regime—for example, by withdrawing funds or shorting equity shares); the bank fails if its fundamental is not strong enough to withstand the aggregate attack. Tests are conducted at uniform intervals starting at the beginning of the distress period. *A bank passes the  $n$ -th test if its remaining fundamental, after accounting for past attacks, is sufficient to withstand a  $k$ -fraction attack by the agents who are yet to act.*<sup>5</sup> We explicitly account for stakeholders’ strategic responses to test results and focus on the worst equilibrium under any given stress test policy.

A stress test with frequency  $N$  divides the unit mass of agents into  $N$  groups, where group  $n$  observes the first  $n$  stress test results before deciding. The stress test results are public. However, agents also observe some noisy private signals. Stress test results affect how agents update their beliefs about the bank’s fundamentals and their higher-order beliefs about others’ behavior (moving now or in the future), which impacts their actions, which in turn affects the future stress test outcomes. We restrict attention to equilibrium in monotone strategies, i.e., an agent is less inclined to attack if he receives more favorable information about the bank’s fundamental. Such an equilibrium results in a unique threshold fundamental such that the bank survives in the end if and only if the fundamental exceeds this *survival threshold*.

Under any stress test policy, there is always an *obedient* equilibrium in which agents ignore their private information and run if and only if the bank has failed a previous test. Therefore, the survival threshold in the obedient equilibrium is the test stringency  $k$ . However, multiple equilibria can emerge in this dynamic coordination game. Our primary interest is the design of *credible stress tests*—*policies under which, even in the worst equilibrium, a bank that passes a test never subsequently fails.* We characterize all *disobedient* equilibria in which the survival threshold fundamental exceeds the test stringency  $k$ . Note that, if such a disobedient equilibrium exists, then the stress test is not credible as the bank may fail after passing the first test according

<sup>4</sup>The excerpt is from [Hirtle and Lehnert \(2015\)](#), which studies the background and design of stress tests as prudential supervision of banks and other large financial institutions.

<sup>5</sup>In practice, standard stress test disclosure policies take this form or are equivalent to it. In the Online Appendix A, we discuss how the disclosures in capital and liquidity stress tests conducted in practice are equivalent to the stress test policy in this paper.

to the worst equilibrium. While, if there exists no disobedient equilibrium, then the worst equilibrium is the obedient one, under which the test results are credible. Therefore, *a policy is credible if and only if there is no such disobedient equilibrium.*

We make three key contributions.

**Characterizing credible policies (Theorem 1).** We provide a necessary and sufficient condition for credibility. We call it **Condition C**. A disobedient equilibrium has two key properties. First, whenever the bank fails a test, all agents attack regardless of their private signals; this causes the bank to fail all subsequent tests, triggering further attacks from all remaining groups and ultimately resulting in bank failure. Second, even if the bank has passed all tests so far, agents in a given group may still attack if their private signal falls below a group-specific information threshold. To characterize a disobedient equilibrium, it therefore suffices to identify the information thresholds ( $\hat{s}_n$ )—below which group- $n$  agents attack even after observing the bank pass all tests up to and including the  $n$ -th test. Because later groups observe more passes, they update their belief about the bank’s survival more favorably and attack less aggressively, so  $\hat{s}_n$  is non-increasing in  $n$ . It follows that any disobedient equilibrium must have *a last disobedient group*  $\bar{n}$ , and the marginal agent (with cutoff signal) in each disobedient group must be indifferent between attacking and not attacking.<sup>6</sup>

*Condition C says that for any candidate disobedient equilibrium in which the first group is the last disobedient group ( $\bar{n} = 1$ ), the marginal agent in that group always strictly prefers not attacking.*

To understand this condition, consider a candidate disobedient equilibrium with  $\bar{n} = 1$ . Group 1 agents attack if their private signal falls below  $\hat{s}_1$ , even after the bank passes the first test. If the bank nonetheless withstands these attacks and passes the second test, all subsequent groups will be obedient, and no further attacks will occur, ensuring the bank’s survival. For the existence of such a disobedient equilibrium, the key object is the *survival belief* of the marginal agent in group 1—the agent with signal  $s_i = \hat{s}_1$  who must be indifferent between attacking and not attacking. If **Condition C** holds, then this indifference cannot hold, meaning a disobedient equilibrium with  $\bar{n} = 1$  does not exist.

**Condition C** is necessary because when it is violated, using continuity, one can find  $\hat{s}_1$ , such that the marginal agent in group 1 is indifferent if all subsequent groups are obedient, and thus, construct a disobedient equilibrium with  $\bar{n} = 1$ , making the stress test non-credible. The sufficient argument rests on a simple observation: under monotone strategies, a stronger bank accumulates less attack than a weaker one, so the difference in their residual fundamentals

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<sup>6</sup>We assume the signal distribution is log-concave, which ensures that if the marginal agent is indifferent, then all agents observing a lower (higher) signal strictly prefer attacking (not attacking).

grows over time. We refer to this as the *widening gap property*. We show that because of this property, the survival belief of the marginal agent in the last disobedient group under  $\bar{n} = 1$  serves as a lower bound on the corresponding belief of the marginal agent in the last disobedient group for any possible disobedient equilibrium. Therefore, if **Condition C** holds, then the lower bound implies that the marginal agent in the last disobedient group strictly prefers not to attack in every candidate disobedient equilibrium—regardless of which group is last—ruling out all such equilibria and establishing credibility.

**Frequency as a policy lever (Theorem 2).** We show that higher test frequency and higher stringency are *substitutes* in achieving credibility. For any frequency  $N$ , a stress test is credible if and only if its stringency exceeds a threshold  $\underline{k}(N)$ , and  $\underline{k}(N)$  is decreasing and discretely convex in  $N$ , as illustrated in Figure 1. The intuition is as follows. Credibility requires convincing the first—and most skeptical—cohort of agents that the bank will survive, *given that the policy will remain equally convincing at every subsequent checkpoint*. The critical quantity for this is the excess attack that the first cohort inflicts beyond what the test certifies:

$$\frac{x - k}{N},$$

where  $x > k$  is the fraction of group-1 agents who attack at the survival threshold. Higher frequency reduces the group size  $1/N$ , limiting the damage the first skeptical cohort can inflict before the next test provides fresh reassurance to later groups; higher stringency raises the floor on what passing reveals. Both shrink the gap between the survival threshold and the passing threshold, making the marginal group-1 agent more optimistic that the bank will clear the next checkpoint and ultimately survive, thereby making **Condition C** easier to satisfy. However, the marginal benefit of additional tests is diminishing:  $\underline{k}(N) - \underline{k}(N + 1)$  decreases with  $N$ . This follows from the quasi-concavity of the marginal agent’s lower bound belief in  $(N, k)$ . We also show that the set of credible policies shrinks—that is, designing credible stress tests becomes harder—when agents are more eager to run on the bank or have more precise private signals; see Figure 2.

**Optimal policy (Theorem 3).** Frequency is an important policy lever because it lowers the ex-ante probability of bank failure. To see this, recall that less stringent tests enable more banks to pass the test, and therefore, a regulator always chooses the least stringent test that is credible. Since  $\underline{k}(N)$  is decreasing, higher frequency allows the regulator to lower the stringency, and still maintain credibility, thereby allowing more banks to pass the tests and survive in the end. However, as test frequency increases, the marginal reduction in failure probability diminishes. Assuming that the administrative cost of conducting more tests is convex, we characterize the optimal frequency of credible stress testing that balances the marginal benefit and

cost. With this characterization established, we examine the policy implications of incorporating frequency in the design of stress tests. For example, regulators typically conduct more frequent tests for systemically important banks. Since preventing the failure of such institutions generates higher social benefit, it is indeed optimal for the social planner to lower the stringency of each test by increasing the testing frequency.

Finally, we conclude by discussing some extensions. We consider the delegation of stress testing to banks, and how banks might set a different optimal frequency than the regulator, because the banks do not internalize the negative externality their failure imposes on the entire system. We also discuss how other regulatory interventions might complement the stress test policy and improve social welfare.

**Related Literature** This paper contributes to the literature on dynamic information design. The regulator in our model is an information designer who, as in [Kamenica and Gentzkow \(2011\)](#), commits to a disclosure rule—here, a sequence of frequent pass/fail stress tests. A growing literature on dynamic information design (see, e.g., [Au, 2015](#); [Ely, 2017](#); [Ely and Szydlowski, 2020](#)) studies how the promise of future disclosure can be used to persuade a single agent over time. Our regulator faces a mass of privately informed agents playing a coordination game. Moreover, disclosed information is *history-dependent*—stress test outcomes depend on the endogenous history of attacks, creating a feedback loop between disclosure and play. Some recent studies (see, e.g., [Ely et al., 2023](#); [Li, Szydlowski and Yu, 2025](#)) have looked into dynamic information disclosure in other strategic contexts. An important challenge in persuading agents playing a coordination game is equilibrium multiplicity: under any disclosure policy, agents may coordinate on equilibria other than the regulator’s preferred one. We follow [Mathevet, Perego and Taneva \(2020\)](#) and [Basak and Zhou \(2025\)](#) in adopting adversarial selection—the regulator designs the policy anticipating that agents will coordinate on the *worst* equilibrium.

The two most closely related papers are [Goldstein and Huang \(2016\)](#) and [Basak and Zhou \(2020\)](#). [Goldstein and Huang \(2016\)](#) pioneered the analysis of stress tests as an information design problem, studying a single test ( $N = 1$ ) in a static bank run game and characterizing the minimum stringency required for credibility. [Basak and Zhou \(2020\)](#) analyzes a dynamic environment similar to ours but focuses exclusively on viability news—public confirmation that the bank remains operational—which corresponds to the zero-stringency case ( $k = 0$ ) and does not require deliberate design. Our paper nests both as special cases: we characterize the full set of credible stress test policies for arbitrary frequency  $N$  and stringency  $k \geq 0$ , establishing the precise trade-off between these two design dimensions and deriving the optimal policy. In doing so, we resolve the theoretical challenge of characterizing all monotone equilibria in a

dynamic coordination game with endogenous public disclosure.

The equilibrium outcome under any credible stress test also connects to the literature on information cascades ([Bikhchandani, Hirshleifer and Welch, 1992](#)): under a credible policy, agents disregard their private signals and follow the public test result, echoing the cascade phenomenon. The key distinction is that cascades in the social learning literature arise from pure informational externalities in the absence of strategic interactions, whereas in our model agents face payoff externalities and strategic complementarities in a dynamic coordination game. Moreover, the cascade-like outcome in our model is an *endogenous design objective*: the regulator deliberately chooses frequency and stringency to make it optimal for agents to act solely on public disclosures. Our work also relates to the broader literature on learning in dynamic coordination environments (see, e.g., [Angeletos, Hellwig and Pavan, 2007](#); [Chamley, 1999](#); [Chassang, 2010](#); [Dasgupta, 2007](#); [Dasgupta, Steiner and Stewart, 2012](#); [Frankel and Pauzner, 2000](#); [Goldstein, Huang and Yang, 2024](#); [Huang, 2017](#); [Kováč and Steiner, 2013](#); [Mathévet and Steiner, 2013](#)); the key difference is that the regulator in our model actively governs the learning process through the design of the disclosure policy.

At a more applied level, our paper contributes to the growing literature on optimal stress test design (see [Goldstein, Sapra et al., 2014](#), for a review). Our central contribution is to introduce test frequency as a policy lever—one that is both practically important and theoretically underexplored. Several recent papers study complementary dimensions of stress test design. [Dai, Luo and Yang \(2024\)](#) considers stress tests for multiple banks and shows that disclosing systemic exposures rather than idiosyncratic shortfalls better mitigates systemic risk. [Inostroza \(2023\)](#) studies regulatory disclosure aimed at persuading multiple audiences—asset market investors and short-term creditors—simultaneously. [Orlov, Zryumov and Skrzypacz \(2023\)](#) highlights a distinct benefit of stress testing: inducing banks to make prudent portfolio choices *ex ante*.<sup>7</sup> Our paper focuses on the persuasion role of frequent disclosure and establishes that frequency and stringency are substitutes in achieving credibility, with concrete implications for how the optimal policy should vary with systemic importance, bank fragility, and macroeconomic conditions.

**Structure** The remainder of the paper is organized as follows. [Section 2](#) presents the model and the stress test policy. Under any stress test policy, [Section 3](#) characterizes the disobedient equilibrium, if such an equilibrium exists. [Section 4](#) identifies the necessary and sufficient condition of credibility such that no disobedient equilibrium can arise. We then demonstrate why higher frequency enhances credibility. [Section 5](#) characterizes the optimal frequent credible stress test, explores its policy implications, and relates our theoretical results to current bank

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<sup>7</sup>Our list of papers is supposed to be representative, but not exhaustive. Other papers that study stress test design include: [Shapiro and Zeng \(2024\)](#), [Quigley and Walther \(2024\)](#), [Parlatore and Philippon \(2022\)](#).

supervisory practice. [Section 6](#) presents some extensions and [Section 7](#) concludes.

## 2 Model

The economy consists of a bank, its stakeholders, and a regulator. An adverse shock may arrive, weakening the bank's asset quality. Stakeholders have incomplete information about the severity of the shock and sequentially decide whether or not to attack. As in a canonical regime change game, the bank fails if it cannot withstand the aggregate attack. Attack can be interpreted differently depending on the context, e.g., creditors run on the bank or increase the collateral requirements for the bank's borrowing, investors refuse to participate in equity issuance or short the bank's equity shares. The regulator conducts frequent stress tests to dissuade the stakeholders from running on the bank.

### 2.1 Basic Setup

**The bank and its stakeholders** An adverse shock arrives at some future date  $T_0$  with probability  $\pi \in (0, 1)$ , and lasts for a fixed duration. If the bank can survive this shock and the attack from stakeholders between  $T_0$  to  $T_1$ , then the bank is "out of the woods." Without loss of generality, we normalize  $T_0$  to 0 and the duration of the distress,  $T_1 - T_0$ , to 1. During the distress period,  $t \in [0, 1]$ , there is a unit mass of stakeholders (agents), indexed by  $i \in [0, 1]$ , who move sequentially: each agent  $i$  decides whether to attack the bank ( $a_i = 1$ ) or not ( $a_i = 0$ ) at time  $t = i$ .

**Deterioration of the bank's fundamental** If the shock arrives, nature selects the bank's post-shock fundamental  $\theta$  from a commonly known prior  $\mathcal{U}[\underline{\theta}, \bar{\theta}]$ . The arrival of this adverse shock is publicly observable but the post-shock bank fundamental  $\theta$  is not. Following the arrival of the adverse shock, the bank's fundamental further deteriorates if agents attack the bank. We model the deterioration of the fundamental as follows.<sup>8</sup> The initial post-shock fundamental is  $\theta^0 = \theta$  and the *residual fundamental* after time  $t \in [0, 1]$  is

$$\theta^t \equiv \theta^0 - w^t = \theta - \int_{i=0}^t a_i \, di, \quad (1)$$

in which  $w^t \in [0, t]$  is the cumulative attack until time  $t$ . If  $\theta^1 = \theta - w^1 \geq 0$ , the bank survives both the exogenous shock and any endogenous attacks (denoted by  $\zeta = P$ ); otherwise, the bank fails (denoted by  $\zeta = F$ ).

Note that the shock can be so severe—i.e.,  $\theta < 0$ —that the bank can fail due to the adverse

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<sup>8</sup>For tractability, we model this deterioration in an abstract way under which the attack negatively impacts the bank's fundamental linearly. The main results from the model should not be bounded to this specific modeling choice.

shock even if no agent attacks. On the other hand, it is also possible that the shock is so mild—i.e.,  $\theta \geq 1$ —that the bank survives even when all agents attack. We call  $[\underline{\theta}, 0)$  the lower dominance region and  $[1, \bar{\theta}]$  the upper dominance region. Moreover, if  $\theta \in [0, 1)$ , the bank will not fail in the absence of attack from the stakeholders. Finally, if there is no adverse shock, then it is commonly known that the bank fundamental is  $\theta \geq \bar{\theta} > 1$ .

**Stakeholders' Incentives** If an agent believes that the bank will (not) fail irrespective of what other agents do, then (not) attacking would be an optimal choice. In this incomplete-information world, as agents are uncertain about the bank's survival, we assume that each agent would not attack if and only if they believe that the bank will survive in the end with a probability at least  $p$ , for some  $p \in (0, 1)$ . The parameter  $p$  captures the willingness to attack. For example, if the bank stakeholders have better outside options for the money they have invested in the bank, then  $p$  is higher, meaning that agents are more inclined to attack.<sup>9,10</sup>

**Exogenous Private Information** Upon the arrival of the adverse shock, each agent receives a noisy private signal regarding the bank's fundamental,

$$s_i = \theta + \sigma \epsilon_i,$$

where the noises  $(\epsilon_i)_i$  are independent across agents conditional on  $\theta$  and identically distributed according to  $\epsilon_i \sim F$ . We assume that  $F$  has support  $[-0.5, 0.5]$ , and it admits a density  $f$ , which is symmetric around 0, and  $f(x) > 0$  for  $x \in [-0.5, 0.5]$  and 0 otherwise. The parameter  $\sigma$  scales the noisiness of the private information. As is standard in the global games literature, we assume  $\underline{\theta} < -\sigma$  and  $\bar{\theta} > 1 + \sigma$ . Under this assumption, the posterior belief of an agent with any private signal  $s_i \in S = [-\frac{\sigma}{2}, 1 + \frac{\sigma}{2}]$  is simply  $F\left(\frac{\theta - s_i}{\sigma}\right)$ .

**Assumption 1.**  $f(\cdot)$  is log-concave.

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<sup>9</sup>To see where the model primitive  $p$  comes from, consider the following example (similar to [Rochet and Vives \(2004\)](#)) in which the creditors (agents) move sequentially in deciding whether or not to roll over the short-term debt. The payoff from withdrawing is 1, regardless of whether the bank ultimately survives or fails. In contrast, the payoff from rolling over is  $r > 1$  if the bank survives, and  $d < 1$  if it fails. Therefore, the payoff difference from rolling over and withdrawing is

$$\Delta u(a_{-i}, \theta) = u(a_i = 0, a_{-i}, \theta) - u(a_i = 1, a_{-i}, \theta) = b \mathbb{1}(\theta^1 \geq 0) - c \mathbb{1}(\theta^1 < 0),$$

where  $b = r - 1 > 0$  is the benefit she gets if the bank survives in the end (promised return  $r$  rather than 1) and  $c = 1 - d > 0$  is the cost she bears if the bank fails in the end (recovery value  $d$  rather than 1). Under this payoff specification, an agent would roll over instead of withdrawing if and only if she believes that the probability of bank survival is at least  $p \equiv \frac{c}{b+c} = \frac{1-d}{r-d} \in (0, 1)$ . It is worth noting that, in practice, the payoff the agents receive when the bank fails can be more nuanced. However, as long as there is a threshold belief  $p \in (0, 1)$  that can convince the agents to roll over, the exact specification plays no role for our main analysis.

<sup>10</sup>Notice that the agents only decide whether to attack or not. We assume that agents cannot trade their claims on the bank's assets among themselves or in any market. In practice, one can think of this assumption as if the market for these claims has broken down in the aftermath of the adverse shock, e.g. the frozen inter-bank markets in 2008.

Under this assumption, the belief of an agent with the higher signal is ranked higher according to the likelihood ratio. Moreover, this ranking is preserved regardless of any public disclosure.<sup>11</sup>

## 2.2 Stress Test: A disclosure policy

The regulator designs a stress test as a *dynamic information disclosure policy*. We study how the disclosure policy influences agents' endogenous attacks and ultimately determines the bank's fate. Throughout the paper, we consider the stress test policy of the following form. The regulator commits to a stress test policy  $\Gamma = (N, k)$ , which is characterized by its frequency  $N \in \mathbb{Z}^+$  and its strength  $k \in [0, 1]$ . More specifically, the policy tests the bank's fundamental  $N$  times at uniform intervals of length  $t_N \equiv \frac{1}{N}$ , and each test features an identical per-capita strength  $k \in [0, 1]$ .

**Definition 1** (Stress Test Policy). *A stress test policy  $\Gamma = (N, k)$  sends a public binary signal  $\zeta_n$  at time  $(n-1)t_N$  for  $n = 1, 2, \dots, N$ . The  $n$ -th test reports at time  $t = (n-1)t_N$  whether the bank's per-capita residual fundamental is greater than  $k$  ("pass",  $\zeta_n = P$ ) or not ("fail",  $\zeta_n = F$ ); that is,*

$$\zeta_n \equiv \begin{cases} P, & \text{if } \frac{\theta^{(n-1)t_N}}{1 - (n-1)t_N} \geq k \\ F, & \text{otherwise.} \end{cases} \quad (2)$$

For a given frequency  $N$ , stress tests are scheduled at times  $t = (n-1)t_N$  for  $n = 1, 2, \dots, N$ .<sup>12</sup> The first test, conducted at time 0, assesses whether the bank could survive if a fraction  $k$  of agents were to attack—specifically, whether the post-shock fundamental  $\theta$  is strong enough to withstand such an attack. This can be interpreted as testing the bank's resilience under a *hypothetical adverse scenario* that weakens its balance sheet and triggers an endogenous attack by  $k$ -fraction of its stakeholders. In other words, after the adverse shock, the bank must retain a sufficient (liquidity or capital) buffer to absorb a  $k$ -fraction attack in order to pass the test. Subsequent tests follow a similar design: each evaluates the bank's ability to withstand a projected adverse scenario in which a  $k$ -fraction attack is launched by the remaining stakeholders who are yet to make their decisions.

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<sup>11</sup>Formally, if  $f(\cdot)$  is log-concave, then for any  $s'' > s'$ ,  $\frac{\text{Prob}(\theta|s'')}{\text{Prob}(\theta|s')} = \frac{f((\theta-s'')/\sigma)}{f((\theta-s')/\sigma)}$  is increasing in  $\theta$  (assuming the denominator is positive). That is, the posterior beliefs are ranked according to likelihood ratio. This implies for any event  $A$ ,  $\frac{\mathbb{P}(\theta|s'', \theta \in A)}{\mathbb{P}(\theta|s', \theta \in A)}$  is also increasing in  $\theta$  (See Theorem 1.4.6. in [Müller and Stoyan \(2002\)](#)).

<sup>12</sup>We follow the convention that the test result for a test conducted at time  $t$  is disclosed before the agent who moves at time  $t$  makes her decision.

### 2.3 Dynamic Coordination Game: Strategies and Equilibrium

Under any policy  $\Gamma = (N, k)$ , the agents play a dynamic coordination game. The agents who move at time  $[(n-1)t_N, nt_N)$  see the same stress test results. We refer to these agents as *group  $n$*  agents, for  $n = 1, 2, \dots, N$ .<sup>13</sup> Notably, the coordination program arises both within each group  $n$  and across different groups of agents.

**Monotone Strategy** A strategy of an agent  $i$  in group  $n$  is a mapping from his private signal ( $s_i$ ) and the first  $n$  stress test results  $\zeta^n$  (public signal) to an action:  $a_i : \mathcal{S} \times \{P, F\}^n \rightarrow \{0, 1\}$ . Throughout this paper, we restrict our attention to monotone strategies. Under a monotone strategy, an agent attacks less when the information (private or public) he receives is more *favorable*. We first clarify what constitutes “more favorable information” in the context of private signals and public results from stress tests. A private signal  $s_i$  is more favorable than  $s'_i$  if and only if  $s_i \geq s'_i$ . For stress test results, we say that a passing grade is strictly more favorable than a failing grade, denoted as  $P \succ F$ ; and, for any group  $n$  agent, we say the public information (test results),  $\zeta^n = \{\zeta_1, \zeta_2, \dots, \zeta_n\}$ , is more favorable than  $\zeta'^n$ , or  $\zeta^n \succeq \zeta'^n = \{\zeta'_1, \zeta'_2, \dots, \zeta'_n\}$ , if and only if  $\zeta_m \succeq \zeta'_m$  for all  $m = 1, \dots, n$ .

**Definition 2** (Monotone Strategy). *A strategy  $a_i$  is a monotone strategy if  $a_i(s'_i, \zeta'^n) \geq a_i(s_i, \zeta^n)$  for any  $(s_i, \zeta^n)$  and  $(s'_i, \zeta'^n)$  such that  $s_i \geq s'_i$  and  $\zeta^n \succeq \zeta'^n$ .*

**Solution Concept** Given any stress test policy  $\Gamma$ , we consider perfect Bayesian equilibrium (PBE) where the agents play symmetric monotone strategies. Henceforth, we refer to this solution simply as *equilibrium*. In equilibrium, each agent maximizes their expected payoff given their beliefs, which must be Bayesian consistent on the equilibrium path. As is typical in coordination games, multiple equilibria may arise. Let  $\mathcal{E}(\Gamma)$  denote the set of equilibria under a given stress test policy  $\Gamma$ , and let  $\mathcal{P}_f(e)$  denote the probability of bank failure under an equilibrium strategy profile  $e \in \mathcal{E}(\Gamma)$ . Under any policy  $\Gamma$ , our focus will be the *worst equilibrium*  $e^W(\Gamma)$ .

**Definition 3** (Worst Equilibrium). *Under a stress test policy  $\Gamma$ , the worst equilibrium  $e^W(\Gamma)$  is the equilibrium with the maximum probability of bank failure*

$$e^W(\Gamma) := \arg \max_{e \in \mathcal{E}(\Gamma)} \mathcal{P}_f(e),$$

*and the maximal probability of bank failure is*

$$\mathcal{P}_f^W(\Gamma) := \mathcal{P}_f(e^W(\Gamma)).$$

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<sup>13</sup>More precisely, agent  $i \in [0, 1)$  belongs to group  $\lfloor iN \rfloor + 1$ , and agent  $i = 1$  belongs to group  $N$ .

## 2.4 The Regulator

The regulator can forecast  $\theta$  through due diligence and observe endogenous attacks over time. She discloses information based on the fundamental  $\theta$  and the history of attacks. However, she does not have access to the private signals received by the agents. Before a potential adverse shock occurs at  $T_0 = 0$ , the regulator commits to a stress test policy  $\Gamma = (N, k)$ . Within the time interval  $[0, 1]$ , she conducts frequent stress tests and publicly discloses the result  $\zeta_n \in \{P, F\}$  at pre-specified dates.

The regulator anticipates that under any stress test policy  $\Gamma$ , the agents will play the worst equilibrium  $e^W(\Gamma)$ . If a bank fails after it passes any stress test, the regulator incurs a tremendous reputational cost because it may undermine the market's confidence in the regulator's competence in screening bad banks from good ones, making future regulatory disclosures ineffective. We assume that the regulator only chooses a *credible* stress test policy.

**Definition 4** (Credible Stress Tests). *A stress test policy  $\Gamma = (N, k)$  is credible, or  $\Gamma \in \Gamma^C$ , if, under the worst equilibrium  $e^W(\Gamma)$ , a bank that passes any of the  $N$  tests never fails.*

The regulator's problem is to choose a credible stress test policy  $\Gamma \in \Gamma^C$  to maximize the social welfare

$$\Lambda(\Gamma) = \left( (1 - \pi) + \pi(1 - \mathcal{P}_f^W(\Gamma)) \right) B - \pi \mathcal{P}_f^W(\Gamma) \chi - C(\Gamma). \quad (3)$$

Recall that the shock arrives with probability  $\pi$ . Under any given policy  $\Gamma$  and based on the worst equilibrium, the bank fails with probability  $\pi \mathcal{P}_f^W(\Gamma)$  and survives with probability  $(1 - \pi) + \pi(1 - \mathcal{P}_f^W(\Gamma))$ . The bank has a charter value of  $B > 0$ , which is forfeited only if the bank fails. Additionally, bank failure incurs a social cost of  $\chi > 0$ .<sup>14</sup> The administrative cost of conducting frequent stress tests,  $C(\Gamma)$ , is independent of the test difficulty  $k$  but increases convexly with the test frequency  $N$ . For simplicity, we denote this cost as  $C(N)$  and it is increasing and convex in  $N$ , meaning that the *marginal cost*, defined as  $\Delta C(N) \equiv C(N + 1) - C(N)$ , is strictly positive and weakly increasing.

## 2.5 Timing of the entire game

We conclude the model description by restating the timing of the game. During normal times, before the potential arrival of an adverse shock, the regulator selects a credible stress test policy  $\Gamma \in \Gamma^C$  to maximize the expected payoff  $\Lambda(\Gamma)$ , anticipating that agents will play the worst-possible equilibrium  $e^W(\Gamma)$  under the chosen policy  $\Gamma$  when the adverse shock arrives. If the shock does not occur, the bank never fails. However, if the shock arrives, the bank—with its

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<sup>14</sup>Note that we focus on microprudential stress tests, but the social cost of failure  $\chi$  serves as a reduced-form representation of the macroprudential implications of stress test design.

post-shock fundamental  $\theta$ —enters a distress period  $t \in [0, 1]$ . The shock’s arrival is common knowledge among stakeholders, who move asynchronously and engage in a dynamic coordination game to decide whether to attack the bank. Stakeholders’ decisions depend on their private signals  $s_i$  about the bank’s fundamental  $\theta$  and the public test results  $\zeta^n$  disclosed under the chosen stress test policy  $\Gamma$ . The bank’s fundamental  $\theta^t$  deteriorates over time due to endogenous attacks, which in turn affects its ability to pass subsequent stress tests and ultimately determines whether it can survive the distress period.

### 3 Equilibrium in the Dynamic Coordination Game

There can be multiple equilibria in the dynamic coordination game under a stress test policy  $\Gamma$ . Characterization of all equilibria is somewhat involved. To provide an intuitive understanding, we start with two benchmarks: (1) no test and (2) a single test. These results are well-known.

**No Test Benchmark:** In this coordination game, agents move asynchronously. However, in the absence of stress tests, agents receive no new public information over time, leaving the strategic environment identical to a simultaneous-move coordination game—asynchronous moves alone do not alter the equilibrium. As in [Morris and Shin \(2003\)](#), a monotone equilibrium takes the form  $a_i(s_i) = \mathbb{1}\{s_i < \hat{s}\}$ : agents attack if and only if their private signal falls below a threshold  $\hat{s}$ . Since a higher fundamental  $\theta$  makes it less likely that agents observe low signals, aggregate attack is decreasing in  $\theta$ , and there exists a unique survival threshold  $\hat{\theta}(\hat{s})$  such that the bank survives if and only if  $\theta \geq \hat{\theta}(\hat{s})$ . The equilibrium threshold  $\hat{s}$  is pinned down by the indifference condition of the marginal agent: the agent with signal  $s_i = \hat{s}$  must be exactly indifferent between attacking and not attacking, which requires her to believe the bank survives with probability exactly  $p$ ,

$$P(\theta \geq \hat{\theta}(\hat{s}) \mid \hat{s}) = p.$$

**Single Test Benchmark:** If there is a single stress test conducted at the beginning, then the equilibrium is exactly as in [Goldstein and Huang \(2016\)](#). Note that passing (failing) a test means  $\theta \geq k$  ( $\theta < k$ ), i.e., if less (more) than  $k$  fraction of agents attack, the bank will survive (fail). This means there is always a monotone equilibrium in which regardless of their private signals, agents do not attack (attack) after then bank passes (fails) the test. We call this the *obedient equilibrium*. However, this may not be the only equilibrium.

Suppose that agents play a monotone strategy:  $a_i(s_i, \zeta) = \mathbb{1}\{s_i < \hat{s}(\zeta)\}$ , where they attack the bank after the bank passes (fails) the test if they observe a signal below  $\hat{s}(P)$  ( $\hat{s}(F)$ ). Let

$$\hat{\mathbf{s}} = \{\hat{s}(P), \hat{s}(F)\}.$$

Since higher  $\theta$ , generate more favorable signals (private and public), there is less attack, and accordingly, there exists  $\hat{\theta}(\hat{s})$  such that the bank survives iff  $\theta \geq \hat{\theta}(\hat{s})$ . If  $\hat{\theta}(\hat{s}) = k$  then we have the obedient equilibrium. There can be disobedient equilibrium with  $\hat{\theta}(\hat{s}) > k$ , in which an agent always attack if the bank fails the test (i.e.,  $\hat{s}(F) = k + \frac{c}{2}$ ), and also attacks even after the bank passes the test if the private signal is not sufficiently high ( $s_i \leq \hat{s}(P)$ ). We simplify this strategy as the cutoff  $\hat{s} = \hat{s}(P)$  below which the agents attack even if they see the bank pass the test. Under this equilibrium, when the fundamental  $\theta < k$ , the bank fails the test and all the agents attack (since they know the bank cannot survive); when  $\theta \in [k, \hat{\theta}(\hat{s}))$ , more agents attack than the bank can withstand and so the bank fails; and when  $\theta \geq \hat{\theta}(\hat{s})$ , less agents attack than the bank can withstand and the bank survives. When the bank passes the test, agents learn that  $\theta \geq k$ , which makes the indifference condition for the marginal agent

$$P(\theta \geq \hat{\theta}(\hat{s}) | \hat{s}, \theta \geq k) = p.$$

If under a stress test policy, such a disobedient equilibrium  $\hat{s}$  with  $\hat{\theta}(\hat{s}) > k$  exists, then the stress test policy is not credible, and if there is no such disobedient equilibrium, then the stress test is credible. [Goldstein and Huang \(2016\)](#) shows that a single stress test is credible iff it is sufficiently stringent because under sufficiently high  $k$ , the above indifference does not hold for any  $\hat{s}$ .<sup>15</sup>

**More than one test:** Suppose the regulator conducts two tests. Two tests results in two groups, while the first group of agents see the first test result, the second group sees both test results. It is easy to check that similar to the case of a single tests, there is an obedient equilibrium in which regardless of their private signals, agents do not attack if the bank passes all the tests so far, and attack, otherwise. There can also be disobedient equilibrium. However, unlike under a single test, the second stress result is *endogenous* since if more agents in group 1 attack, then the bank might not have sufficient residual fundamental to pass the second test. Therefore, along with the payoff externality across groups, there is also an information externality.

For a general stress test policy  $\Gamma = (N, k)$ , there are  $N$  groups. Suppose agents follow a monotone strategy:  $a_i(s_i, \zeta^n) = \mathbb{1}\{s_i < \hat{s}_n(\zeta^n)\}$ , where  $\zeta^n$  is any possible sequence of test results that group  $n = 1, 2, \dots, N$  might observe. We write this monotone strategy as

$$\hat{s} = (\hat{s}_n)_{n=1,2,\dots,N}, \text{ where } \hat{s}_n = (\hat{s}_n(\zeta^n))_{\zeta^n \in \{P,F\}^n}.$$

Since there can be many sequences of pass/fail results under frequent tests, what agents learn from these test results may be unclear, making the equilibrium characterization difficult. Nev-

<sup>15</sup>If  $\hat{\theta}(\hat{s}) < k$ , we have a disobedient equilibrium in which agents do not attack after the bank passes a test, and even after it fails they do not attack unless private signals are sufficiently low. Since, we are interested in the worst equilibrium and there is always an obedient equilibrium with  $\hat{\theta}(\hat{s}) = k$ , we can ignore such disobedient equilibrium.

ertheless, following three simple steps, we characterize the equilibrium. The steps are as follows.

Step 1: We show that an equilibrium  $\hat{\mathbf{s}}$  results in  $(N + 1)$  fundamental cutoffs  $(\underline{\theta}_n(\hat{\mathbf{s}}))_{n=0}^N$ , one for passing each of the  $N$  tests, and one for surviving in the end  $\underline{\theta}_N(\hat{\mathbf{s}}) = \hat{\theta}(\hat{\mathbf{s}})$ .

Step 2: There is always an obedient equilibrium, in which agents ignore their private signals and follow the stress test result, which leads to  $\hat{\theta}(\hat{\mathbf{s}}) = k$ .

Step 3: In a disobedient equilibrium with  $\hat{\theta}(\hat{\mathbf{s}}) > k$ , if one exists, agents always attack whenever the bank has failed a previous test; even if the bank has passed all tests so far, they may still attack if their private signal is below some cutoff. Moreover, there is a last disobedient group because agents in later groups attack less aggressively than earlier groups after they see more pass results.<sup>16</sup>

Equipped with these three results, we then characterize a disobedient equilibrium, under which the marginal agent in any disobedient group must be indifferent between attacking and not attacking.

### 3.1 Step 1: Passing thresholds are survival threshold

**Lemma 1** (Fundamental Thresholds). *Under any stress test policy  $\Gamma = (N, k)$ , if agents play some monotone strategy profile  $\hat{\mathbf{s}}$ , there exist  $N$  fundamental thresholds  $\underline{\theta}_n(\hat{\mathbf{s}})$  such that the bank passes the  $(n + 1)$ th test if and only if  $\theta \geq \underline{\theta}_n(\hat{\mathbf{s}})$  for  $n = 0, 1, \dots, N - 1$ . Moreover, there exists a unique survival threshold  $\hat{\theta}(\hat{\mathbf{s}})$  such that the bank survives if and only if  $\theta \geq \hat{\theta}(\hat{\mathbf{s}})$ .*

Intuitively, when agents follow monotone strategies, a bank with a stronger post-shock fundamental  $\theta' > \theta$  will always pass a test if a bank with  $\theta$  does. Conversely, a bank with a weaker fundamental  $\theta' < \theta$  will fail the same test if a bank with  $\theta$  does. This monotonicity allows us to define the fundamental threshold  $\underline{\theta}_n$ , which determines the outcome of the  $(n + 1)$ th test. Following the same logic, there always exists a *survival threshold*,  $\hat{\theta}$ , that dictates whether the bank ultimately survives the distress period. The survival threshold can be interpreted as the fundamental threshold to pass the  $(N + 1)$ th test (with no more agents left to make decision) and we use  $\underline{\theta}_N$  and  $\hat{\theta}$  interchangeably.

It follows from the above lemma that the ex-ante probability of bank failure under any equilibrium  $e \in \mathcal{E}(\Gamma)$  is given by:

$$\mathcal{P}_f(e) = \frac{\hat{\theta}(e) - \theta}{\bar{\theta} - \theta}. \quad (4)$$

Accordingly, we can compare the resulting survival thresholds  $\hat{\theta}(e)$  to determine the worst

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<sup>16</sup>Formally, a group is disobedient if they attack even after seeing the bank has passed all the past tests.

equilibrium:

$$e^W(\Gamma) := \arg \max_{e \in \mathcal{E}(\Gamma)} \hat{\theta}(e).$$

### 3.2 Step 2: Obedience and credibility

Consider the following strategy:  $a_i(s_i, \zeta^n) = \mathbb{1}\{\zeta^n \neq \{P, \dots, P\}\}$ , i.e., agents ignore their private signal and attack the bank whenever it fails a previous test, and not attack otherwise. Given a stress test policy  $\Gamma$ , an equilibrium is *obedient*, denoted by  $e^o(\Gamma)$ , if the agents follow the above strategy.

**Lemma 2** (Obedient Equilibrium). *Under any stress test policy  $\Gamma = (N, k)$ , obedient equilibrium exists, i.e.,  $e^o(\Gamma) \in \mathcal{E}(\Gamma)$ . Additionally, for any policy  $\Gamma = (N, k)$ , the equilibrium survival threshold  $\hat{\theta}(e) = k$  if and only if the agents play the obedient equilibrium  $e = e^o(\Gamma)$ .*

Recall that under a single stress test policy  $N = 1$ , there is an obedient equilibrium. The same argument can be extended for any  $N > 1$ . Consider an agent  $i$  in say a group  $n$ . If all other agents follow the obedient strategy, then an agent who observes the bank passing all previous tests, expects it to pass the next test. This means that agents in the next group will also not attack, and therefore, the bank will pass the following test, and so on, and the bank survives. Therefore, she will not attack. On the other hand, if the bank has failed a previous test, she expects all the agents in her group to attack, which means the bank cannot pass the next test. This means all the agents in the next group will also attack, and it will not pass the following tests, and so on, and finally fail. Therefore, she will attack.

When agents play the obedient strategy, on path, a bank with  $\theta \geq k$  passes all the tests and survives, and a bank with  $\theta < k$  fails all the tests and fails in the end, i.e.,  $\hat{\theta}(e^o(\Gamma)) = k$ . Finally, if  $\hat{\theta}(e(\Gamma)) = k$ , then all agents will always attack (never attack) if the bank fails (passes) the first test since that test result implies that  $\theta < \hat{\theta} = k$  ( $\theta \geq \hat{\theta} = k$ ), regardless of their private signals. Hence, the bank passes the second test if and only if  $\theta \geq \hat{\theta} = k$ . This implies that, similar to agents' response to the first test result, the agents in equilibrium must follow the test results in the second and subsequent tests. Therefore,  $\hat{\theta}(e(\Gamma)) = k$  implies that agents play the obedient equilibrium,  $e(\Gamma) = e^o(\Gamma)$ .

Typically, there are multiple equilibria and we are interested in the worst equilibrium. Since under any  $\Gamma$ , the obedient equilibrium  $e^o(\Gamma)$  always exists and results in  $\hat{\theta}(e^o(\Gamma)) = k$ , the worst equilibrium  $e^W(\Gamma)$  must be such that  $\hat{\theta}(e^W(\Gamma)) \geq \hat{\theta}(e^o(\Gamma)) = k$ . Recall that the equality holds iff  $e^W(\Gamma) = e^o(\Gamma)$ . Let us define the set of equilibrium

$$\mathcal{E}^D(\Gamma) \equiv \{e \in \mathcal{E}(\Gamma) | \hat{\theta}(e) > k\}.$$

For any  $\Gamma$ , if the worst equilibrium is not the obedient one, then it must hold that  $e^W(\Gamma) \in$

$\mathcal{E}^D(\Gamma)$ . We refer to any equilibrium in  $\mathcal{E}^D(\Gamma)$  as *disobedient equilibrium*.<sup>17</sup>

**Proposition 1** (Credibility and Obedience). *A stress test policy  $\Gamma$  is credible (i.e.,  $\Gamma \in \Gamma^C$ ) if and only if there is no disobedient equilibrium under  $\Gamma$ , that is,*

$$\mathcal{E}^D(\Gamma) = \emptyset.$$

If  $\mathcal{E}^D(\Gamma) \neq \emptyset$ , then the worst equilibrium features  $\hat{\theta}(e^W(\Gamma)) > k$ . According to the definition of credible stress tests (see definition 4), such a policy is not credible because there exists  $\theta \in (k, \hat{\theta}(e^W(\Gamma)))$  such that a bank with this post-shock fundamental would pass the first test (i.e.,  $\theta \geq k$ ) but ultimately fail (i.e.,  $\theta < \hat{\theta}(e^W(\Gamma))$ ). Conversely, if the stress test policy  $\Gamma$  satisfies  $\mathcal{E}^D(\Gamma) = \emptyset$ , then the worst equilibrium under this policy is the obedient equilibrium, i.e.,  $e^W(\Gamma) = e^o(\Gamma)$ . Under this equilibrium, the bank passes all tests and survives the distress period if and only if  $\theta \geq k$ , implying that the stress test policy is credible.

### 3.3 Step 3: Properties of disobedient equilibrium

**Lemma 3** (Properties of disobedient equilibrium). *Given a stress test policy  $\Gamma = (N, k)$ , any equilibrium  $e(\Gamma) \in \mathcal{E}^D(\Gamma)$ , if exists, must have the following properties*

1. *a player  $i$  in any group  $n$ , always attacks if the bank fails a previous test, and even if the bank passes all the tests so far, she may still attack if  $s_i < \hat{s}_n$ .*
2. *there is a last disobedient group  $\bar{n} \in \{1, 2, \dots, N\}$  in which agents attack even after the bank passes all the past tests, and*

$$k = \underline{\theta}_0 < \underline{\theta}_1 < \dots < \underline{\theta}_{\bar{n}} = \underline{\theta}_{\bar{n}+1} = \dots = \underline{\theta}_N = \hat{\theta}.$$

We show that in a monotone equilibrium, agents must attack more (less) aggressively when they see an extra fail (pass) stress test result. This makes the pass thresholds non-decreasing:

$$k = \underline{\theta}_0 \leq \underline{\theta}_1 \leq \dots \leq \underline{\theta}_{N-1} \leq \underline{\theta}_N = \hat{\theta}$$

To see why, suppose they are not. Then we can find an  $n$  and a  $\theta$  such that when the fundamental is  $\theta$ , the bank will pass the first  $(n - 1)$  tests, fail the  $n$ -th test, and then pass the  $(n + 1)$ -th test. This can happen only if the proportion of attack from group  $(n - 1)$  is strictly above  $k$  and that from group  $n$  is strictly below  $k$  when the fundamental is  $\theta$ . However, note that compared to group  $(n - 1)$ , group  $n$  agents see an extra fail stress test result. This gives us a contradiction. These non-decreasing thresholds imply that, in any such equilibrium, once a bank fails a test, it cannot pass any subsequent test and therefore must ultimately fail.

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<sup>17</sup>A disobedient equilibrium could also result in  $\hat{\theta} < k$  if some agents do not attack even when the bank fails a test. However, we ignore such an equilibrium since the worst equilibrium will always have  $\hat{\theta} \geq k$ .

If an agent  $i$  in any group  $n$  observes that the bank has failed a previous test  $m \leq n$ , he learns that  $\theta < \underline{\theta}_{m-1} \leq \hat{\theta}$ , which means he understands that the bank will not survive in the end, and therefore, he attacks. If the bank has passed all the tests so far, he may still attack if his private signal is not sufficiently high, i.e.,  $s_i < \hat{s}_n$ . Therefore, the equilibrium strategy can be succinctly represented by  $N$  information thresholds

$$\hat{\mathbf{s}} = (\hat{s}_n)_{n=1}^N.$$

Since agents attack less aggressively after observing more passes,  $\hat{s}_n$  is non-increasing in  $n$ . It follows that if group  $n$  is obedient, all subsequent groups must also be obedient—that is, there exists a last disobedient group  $\bar{n}$ . To see why the passing thresholds  $\underline{\theta}_n$  are strictly increasing up to  $\bar{n}$ , consider any disobedient group  $n \leq \bar{n}$  and suppose the fundamental is exactly at the passing threshold,  $\theta = \underline{\theta}_{n-1}$ . By definition, a bank with this fundamental just passes the first  $n$  tests. Since group  $n$  is disobedient, more than a fraction  $k$  of its agents attack at  $\theta = \underline{\theta}_{n-1}$ , leaving the bank with insufficient strength to pass the  $(n+1)$ th test, so  $\underline{\theta}_n > \underline{\theta}_{n-1}$ . The alternative  $\underline{\theta}_n \leq \underline{\theta}_{n-1}$  leads to a contradiction: passing the first  $n$  tests would then guarantee passing the  $(n+1)$ th test; and since all later groups attack even less aggressively, the bank would pass all subsequent tests and survive—giving group  $n$  no incentive to attack in the first place. Therefore,  $\underline{\theta}_n$  is strictly increasing in  $n$  for all  $n \leq \bar{n}$ , and remains constant at  $\hat{\theta}$  thereafter.

### 3.4 Characterizing Disobedient Equilibrium

Equipped with the three results, we now characterize a disobedient equilibrium by specifying the conditions  $\hat{\mathbf{s}} = (\hat{s}_n)_{n=1}^N$  must satisfy.

**Proposition 2** (Characterization of Disobedient Equilibrium). *Under any stress test policy  $\Gamma$ , if a disobedient equilibrium  $\hat{\mathbf{s}} = (\hat{s}_n)_{n=1}^N$  exists, then it must satisfy the following conditions.*

1. *There exists a last disobedient group  $\bar{n}(\hat{\mathbf{s}}) \in \{1, 2, \dots, N\}$  and accordingly, the bank survives if it passes the first  $\bar{n}(\hat{\mathbf{s}}) + 1$  tests, i.e.,  $k = \underline{\theta}_0 < \underline{\theta}_1 < \dots < \underline{\theta}_{\bar{n}} = \underline{\theta}_{\bar{n}+1} = \dots = \underline{\theta}_N = \hat{\theta}$ .*
2. *For any disobedient group  $n \leq \bar{n}(\hat{\mathbf{s}})$ , the marginal agent  $\hat{s}_n$  is indifferent between attacking and not attacking:*

$$\mathbb{P}(\theta \geq \hat{\theta} \mid \zeta^n = \{P, \dots, P\}, \hat{s}_n) = \frac{F\left(\frac{\hat{s}_n - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_n - \underline{\theta}_{n-1}}{\sigma}\right)} = p, \quad (I_n)$$

*where the fundamental threshold  $\underline{\theta}_{n-1}$  is such that the residual strength is just enough to pass the*

$n$ -th test

$$\theta^{\frac{n-1}{N}}(\theta = \underline{\theta}_{n-1}) = \underline{\theta}_{n-1} - \frac{1}{N} \sum_{m=1}^{n-1} F\left(\frac{\hat{s}_m - \underline{\theta}_{n-1}}{\sigma}\right) = k \left(1 - \frac{n-1}{N}\right). \quad (F_n)$$

3. For any obedient group  $n > \bar{n}(\hat{\mathbf{s}})$ , the agents do not attack if the bank passes the first  $\bar{n}(\hat{\mathbf{s}}) + 1$  tests, i.e.,  $\hat{s}_n = \underline{\theta}_{\bar{n}} - \frac{\sigma}{2}$ .

Consider any disobedient group ( $n \leq \bar{n}$ ). After observing that the bank passes all previous tests, an agent  $i$  in this group believes that  $\theta \geq \underline{\theta}_{n-1}$ . Note that unlike under a single test, the threshold the passing threshold  $\underline{\theta}_{n-1}$  depends on the strategy of the earlier groups (see  $(F_n)$ ). Since agents are not obedient, the bank may not survive even after passing all the tests so far. An agent who receives a private signal  $s_i$  believes that the bank will survive with probability

$$\mathbb{P}(\theta \geq \hat{\theta} | s_i, \theta \geq \underline{\theta}_{n-1}) = \frac{F\left(\frac{s_i - \hat{\theta}}{\sigma}\right)}{F\left(\frac{s_i - \underline{\theta}_{n-1}}{\sigma}\right)}.$$

It follows from Assumption 1 that this belief is increasing in  $s_i$ . In equilibrium, the marginal agent  $\hat{s}_n$  must be indifferent between attacking and not attacking, i.e., believe that the probability of survival is  $p$  (see  $(I_n)$ ). This ensures, that an agent with signal  $s_i < \hat{s}_n$  strictly prefers attacking and an agent with signal  $s_i > \hat{s}_n$  strictly prefers not attacking.

## 4 Credible Stress Test Design

Next, we characterize the set of credible stress test policies,  $\Gamma^C$ , and discuss the impact of frequency as a policy lever in the design of credible stress test policy. We also discuss what helps or hurts credibility.

### 4.1 Credible Stress Tests

By Proposition 1, a stress test policy  $\Gamma = (N, k)$  is credible (i.e.,  $\Gamma \in \Gamma^C$ ) if and only if, under this policy, there does not exist any disobedient equilibrium  $\mathcal{E}^D(\Gamma) = \emptyset$ . By identifying the necessary and sufficient condition for  $\mathcal{E}^D(\Gamma) = \emptyset$ , the following theorem characterizes the set of credible stress test policies  $\Gamma^C$ .

**Theorem 1** (Credible Tests). *The set of credible stress tests is*

$$\Gamma^C = \left\{ (N, k) \left| \underbrace{\min_{x \in (k, 1]} \frac{x}{F\left(F^{-1}(x) + \frac{x-k}{N\sigma}\right)}}_{\text{Condition C}} > p \right. \right\}. \quad (5)$$

We proceed by outlining the main argument. First, consider a disobedient equilibrium in

which the first group is the last disobedient group ( $\bar{n} = 1$ ). We define

$$G(x, N, k) := \frac{x}{F\left(F^{-1}(x) + \frac{x-k}{N\sigma}\right)}, \quad (\text{G})$$

which is the belief of the marginal agent in group 1 that the bank will survive when (i) group 1 is the last disobedient group ( $\bar{n} = 1$ ), and (ii) the fraction of attack from group 1 at  $\theta = \hat{\theta}$  is  $x$ . We then use a *widening gap property*: for any two values of fundamental  $\theta$ , say  $\theta'$  and  $\theta''$  with  $\theta'' > \theta'$ , the gap between the residual fundamentals widens over time because under a monotone strategy, a bank experiences less attack when  $\theta$  is higher. This property gives us the following inequality

$$\theta'' - \theta' \leq \theta^t(\theta = \theta'') - \theta^t(\theta = \theta'). \quad (\text{W})$$

Using this widening gap property, we show that  $G(x, N, k)$  serves as a lower bound of the belief (that the bank will survive) of the marginal agent in any last disobedient group  $\bar{n} (> 1)$ , when the fraction of attack from the last disobedient group at  $\theta = \hat{\theta}$  is  $x$ . We explain why  $G(\cdot)$  serves as a lower bound after we demonstrate why **Condition C** is necessary and sufficient for credibility.

Suppose that **Condition C** is violated. Then there exists  $x \in (k, 1]$  such that  $G(x, N, k) \leq p$ . Since  $G(k, N, k) = G(1, N, k) = 1$ , by continuity, there exists  $x^* \in (k, 1]$  for which  $G(x^*, N, k) = p$ . We can then construct a disobedient equilibrium in which the first group is the last disobedient group ( $\bar{n} = 1$ ), the fraction of attack from the first group at the survival threshold is  $x^*$ , and the marginal agent  $\hat{s}_1$  in group 1 is indifferent:  $G(x^*, N, k) = p$ . Therefore,  $\mathcal{E}^D(\Gamma) \neq \emptyset$ . This makes **Condition C** a necessary condition for credibility.

Next, suppose that **Condition C** is satisfied. This means for all  $x \in (k, 1]$ ,  $G(x, N, k) > p$ . Therefore, if at the survival threshold of fundamental  $\hat{\theta}$ ,  $x$  fraction of agents attack from the last disobedient group  $\bar{n}$ , then the marginal agent in the last disobedient group strictly prefers not attacking. This rules out any disobedient equilibrium; i.e.,  $\mathcal{E}^D(\Gamma) = \emptyset$ . This makes **Condition C** a sufficient condition for credibility.

**Why does  $G(\cdot)$  serve as a lower bound?** To understand (G), note that when  $\bar{n} = 1$ , if the bank can survive the attack from group 1 and pass the second test, it will survive, i.e.,  $\underline{\theta}_1 = \hat{\theta}$ . Therefore, the survival threshold  $\hat{\theta}$  is such that (see ( $F_n$ ))

$$\hat{\theta} = \frac{x}{N} + k \left(1 - \frac{1}{N}\right), \text{ where } x = F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right)$$

is the fraction of attack from group 1 when  $\theta = \hat{\theta}$ . In a disobedient equilibrium  $\hat{\theta} > k$ , which means fraction of attack at the survival threshold  $x > k$ . Consider the marginal agent  $\hat{s}_1$  in group 1. When the bank passes the first test, he learns  $\theta \geq k$ . However, this is not sufficient for

survival. For the bank to survive, it must be that  $\theta \geq \underline{\theta}_1 = \hat{\theta}$ , so that the bank can withstand the attack from group 1, and pass the second test. Therefore, she believes that the bank will survive with probability

$$\mathbb{P}(\theta \geq \hat{\theta} \mid \hat{s}_1, \theta \geq k) = \frac{F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_1 - k}{\sigma}\right)} = \frac{F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma} + \frac{\hat{\theta} - k}{\sigma}\right)} = \frac{x}{F\left(F^{-1}(x) + \frac{x - k}{N\sigma}\right)} = G(x, N, k).$$

Next, consider a disobedient equilibrium in which the last disobedient group  $\bar{n} > 1$ . After seeing that the bank has passed the first  $\bar{n}$  tests, the agents learn that  $\theta \geq \underline{\theta}_{\bar{n}-1}$ . By definition, when  $\theta = \underline{\theta}_{\bar{n}-1}$ , the residual fundamental is just enough to withstand  $k$  fraction of attack from the agents who are yet to make their decisions, i.e.,

$$\theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}-1}) = \left(1 - \frac{\bar{n} - 1}{N}\right)k.$$

However, this is not sufficient for survival. If the bank can withstand the attack from group  $\bar{n}$  and pass the next test, then it survives, i.e.,  $\underline{\theta}_{\bar{n}} = \hat{\theta}$ . By definition, when  $\theta = \underline{\theta}_{\bar{n}}$ , the residual fundamental is just enough to sustain the attack from the  $\bar{n}$ -th group and then pass the next test (if there is one); that is,

$$\theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}}) = \frac{x}{N} + \left(1 - \frac{\bar{n}}{N}\right)k.$$

Therefore, the distance between the two residual fundamentals is

$$\theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}}) - \theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}-1}) = \frac{x - k}{N}.$$

It then follows from the widening gap property (W) that

$$\underline{\theta}_{\bar{n}} - \underline{\theta}_{\bar{n}-1} \leq \theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}}) - \theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}-1}) = \frac{x - k}{N}. \quad (6)$$

When  $\bar{n} = 1$ , the distance between the threshold to survive and the threshold to pass the first test is  $\frac{x-k}{N}$ . That is, the above holds with equality. When  $\bar{n} > 1$ , for any fraction of aggregate attack from the last disobedient group  $x \in (k, 1]$ , the gap between the threshold for survival  $\underline{\theta}_{\bar{n}}$  and the threshold for passing all the tests so far  $\underline{\theta}_{\bar{n}-1}$  is lower than the case with  $\bar{n} = 1$ . It follows from (6) that the marginal agent in the last disobedient group is more optimistic about the bank's survival, i.e.,

$$\mathbb{P}(\theta \geq \hat{\theta} \mid \hat{s}_{\bar{n}}, \theta \geq \underline{\theta}_{\bar{n}-1}) = \frac{F\left(\frac{\hat{s}_{\bar{n}} - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_{\bar{n}} - \underline{\theta}_{\bar{n}-1}}{\sigma}\right)} = \frac{F\left(\frac{\hat{s}_{\bar{n}} - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_{\bar{n}} - \hat{\theta}}{\sigma} + \frac{\hat{\theta} - \underline{\theta}_{\bar{n}-1}}{\sigma}\right)} \geq \frac{x}{F\left(F^{-1}(x) + \frac{x - k}{N\sigma}\right)} = G(x, N, k),$$

which gives us the lower bound.

## 4.2 Frequency and Credibility

In this section, we investigate how frequency helps credibility. It follows from [Basak and Zhou \(2020\)](#) that a stress test with  $k = 0$  is credible when it is sufficiently frequent. Proposition 3 shows that this result holds true for any stringency  $k$ .

**Proposition 3** (Sufficiently frequent tests are credible). *A stress test  $\Gamma = (N, k)$  is credible if it is sufficiently frequent, i.e.,  $N \geq \underline{N}(k)$ , where*

$$\underline{N}(k) \equiv \min \left\{ N \in \mathbb{Z}_+ \left| \min_{x \in (k, 1]} G(x, N, k) > p \right. \right\}. \quad (7)$$

Since  $G(x, N, k)$  is increasing in  $N$  and converges to 1 when  $N \rightarrow \infty$ , there must exist  $\underline{N}(k)$  such that when  $N \geq \underline{N}(k)$ , the necessary and sufficient condition for credibility, [Condition C](#), will be satisfied. To understand why  $G(x, N, k)$  increases with  $N$ , recall that  $G(x, N, k)$  is the belief of the marginal agent in group 1 when (i) the first group is the last disobedient group, i.e.,  $\bar{n} = 1$ , and (ii) the fraction of attack from group 1 at  $\theta = \hat{\theta}$  is  $x$ . After the bank passes the first test, agents learn that the bank can survive if less than  $k$  fraction of agents attack, i.e.,  $\theta \geq \underline{\theta}_0 = k$ . However, passing the first test does not guarantee survival; to survive  $\theta \geq \hat{\theta} = \underline{\theta}_1$ . Only then the bank can withstand the endogenous attack from the first group  $x$  (which exceeds fraction  $k$ ), and pass the second test, i.e., can still withstand  $k$  fraction of attack from the rest of the groups. Therefore, as we can see in [\(A.9\)](#), the distance between the threshold for survival ( $\hat{\theta} = \underline{\theta}_1$ ) and the threshold for passing the first test ( $\underline{\theta}_0 = k$ ) is

$$\hat{\theta} - \underline{\theta}_0 = \frac{x - k}{N}.$$

This distance is the same as the aggregate attack from group 1 (the last disobedient group) at  $\theta = \hat{\theta}$  in excess of  $k$ . Since under higher frequency  $N$ , each group is smaller (size  $\frac{1}{N}$ ), the excess attack is lower. The lower this excess attack, the smaller the distance between  $\hat{\theta}$  and  $\underline{\theta}_0$ , which makes the marginal agent more optimistic about the survival of the bank after observing that the bank has passed the first test, i.e.,  $G(x, N, k)$  increases in  $N$ , which gives us the above result.

Note that  $G(x, N, k)$  also increases in  $k$  because a higher stringency  $k$  decreases the above excess attack for any  $x$ . Moreover,  $G(x, N, k)$  converges to 1 as  $k \rightarrow 1$ , which means that there is a threshold  $\underline{k}(N)$  such that a stress test is credible when  $k > \underline{k}(N)$ . This generalizes the result in [Goldstein and Huang \(2016\)](#) which establishes the same for  $N = 1$ . More importantly, our analysis not only generalizes the two existing results, [Theorem 2](#) shows the connection between the two, i.e., how a higher frequency impacts the minimum stringency  $\underline{k}(N)$  needed for credibility.

**Theorem 2** (Higher frequency aids credibility). *A stress test  $\Gamma = (N, k)$  is credible if  $k > \underline{k}(N)$ , where*

$$\underline{k}(N) = \inf \left\{ k \in [0, 1] : \min_{x \in (k, 1]} G(x, N, k) > p \right\}, \quad (8)$$

*and  $\underline{k}(N)$  is decreasing and discretely convex in  $N$  until  $\underline{k}(N)$  becomes 0, and remains 0 thereafter.*

To explain how more frequent tests help credibility, we first define

$$y(N, k) := \min_{x \in (k, 1]} G(x, N, k).$$

Then, the set of credible stress tests simplifies to  $\Gamma^C = \{(N, k) | y(N, k) > p\}$ .<sup>18</sup> Since  $G(x, N, k)$  is increasing in  $k$  and  $N$  for any  $x \in (k, 1]$ , so is  $y(N, k)$ . Therefore, when the regulator conducts an additional test, she can make the tests less stringent and still maintain credibility, i.e., satisfy the necessary and sufficient condition for credibility—**Condition C**. This makes  $\underline{k}(N)$  decreasing.

Frequency is an important policy lever in the design of stress tests because it lowers the minimum stringency  $\underline{k}(N)$  needed for credibility. A lower stringency  $k$  enables more banks to pass the tests, and if  $k > \underline{k}(N)$  then the stress test is credible, and any bank that passes a test never fails, i.e.,  $\hat{\theta}(e^W(\Gamma)) = k$ . Therefore, conducting more tests allows the regulator to make the tests easier and still be credible, which lowers the ex ante probability of bank failure in the worst equilibrium  $P_f^W(\Gamma) = \frac{k - \underline{\theta}}{\theta - \underline{\theta}}$ .

However, the value of frequency as a policy lever diminishes as the regulator conducts more tests. Note that  $\underline{k}(N)$  is discrete convex, i.e.,  $\underline{k}(N) - \underline{k}(N + 1)$  decreases with  $N$ . This arises because  $G(x, N, k)$  is quasi-concave in  $(N, k)$ , which makes  $y(N, k)$  quasi-concave in  $(N, k)$ , which in turn means that the upper contour set, i.e.,  $(N, k)$  such that  $y(N, k) > p$  is convex.

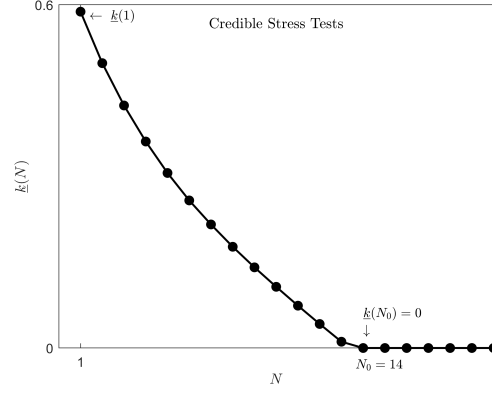
**Figure 1** provides a graphical illustration of  $\underline{k}(N)$  for specific parameter values. The result in [Goldstein and Huang \(2016\)](#) is the extreme case with  $N = 1$  and the result in [Basak and Zhou \(2020\)](#) is the extreme case with  $\underline{k}(N) = 0$ .<sup>19</sup>

### 4.3 What helps or hurts Credibility?

Thus far, we have identified the set of credible stress test policies,  $\Gamma^C$ , and established the properties of the required test strength,  $\underline{k}(N)$ , under any test frequency  $N$  for ensuring credibility.

<sup>18</sup>Formally, we extend  $G(x, N, k)$  over the domain  $x \in [0, 1]$  by defining  $\tilde{G}(x, N, k) = G(x, N, k) \mathbb{1}(x \in (k, 1]) + \mathbb{1}(x \in [0, k])$ , which means  $y(N, k) = \min_{x \in [0, 1]} \tilde{G}(x, N, k)$ .

<sup>19</sup>Instead of considering a stress test design, [Basak and Zhou \(2020\)](#) focus on how an asynchronous debt structure mitigates inefficient coordination risk. They show that when the debt structure is sufficiently asynchronous (corresponding to  $N \geq \underline{N}(0)$  in our framework), the borrower's continued viability alone is sufficient to persuade all agents to roll over maturing debt—that is,  $\underline{k}(N) = 0$  for all  $N \geq \underline{N}(0)$ .

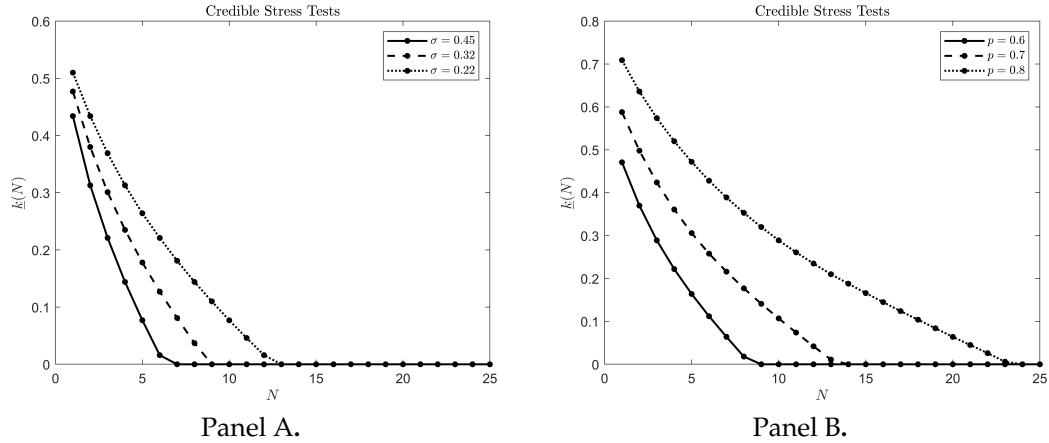


**Figure 1.**  $\underline{k}(N)$  - The strength required to make the  $N$  times repeated stress tests credible.

*Note:* The figure represents the relationship between minimum stringency for a credible test  $\underline{k}(N)$  and the frequency  $N$ . The density of private signal noise  $\epsilon$  is  $f(x) = (2 + 4x) \cdot \mathbf{1}(-0.5 \leq x < 0) + (2 - 4x) \cdot \mathbf{1}(0 \leq x \leq 0.5)$ . The value of parameters  $(p, \sigma) = (0.7, 0.33)$ .

Next, we discuss how the model primitives influence the set of credible stress tests. We find that it is easier for a frequent stress test policy to be credible when the private information is less informative or when the agents' willingness to attack is lower.

**Proposition 4.** *The set of credible stress test policies,  $\Gamma^C$ , expands as  $\sigma$  increases and as  $p$  decreases.*



**Figure 2.** Impact of  $p$  and  $\sigma$  on  $\Gamma^C$

*Note:* Specification:  $f(x) = (2 + 4x) \cdot \mathbf{1}(-0.5 \leq x < 0) + (2 - 4x) \cdot \mathbf{1}(0 \leq x \leq 0.5)$ . Panel A illustrates how the minimum stringency for credibility,  $\underline{k}(N)$ , varies with test frequency  $N$  for different values of the private signal noise parameter  $\sigma$ . Panel B illustrates the same relationship for different values of creditors' willingness to withdraw  $p$ .

Figure 2 illustrates this result. The necessary and sufficient condition for credibility (**Condition C**) requires that  $G(x, N, k) > p$  for all  $x \in (k, 1]$ . Clearly, a lower  $p$  makes it easier to satisfy the above condition. On the other hand, a higher  $\sigma$  means the private signals are noisier. Accordingly, the positive stress test result is more effective in persuading the marginal agent

in group 1, when group 1 is the last disobedient group, i.e.,  $G(x, N, k)$  is higher for higher  $\sigma$ . Therefore, lower  $p$  or higher  $\sigma$  expands  $\Gamma^C$ , i.e., makes it easier to design credible stress tests.

## 5 Optimal Stress Test Frequency

Next, we characterize the optimal choice of the regulator. She anticipates that under any stress test policy  $\Gamma$ , the agents will play the worst equilibrium  $e^W(\Gamma)$ . She chooses a credible stress test policy  $(N, k) \in \Gamma^C$  that ensures that in the worst equilibrium, a bank that passes a test never fails. She maximizes the social welfare under the worst equilibrium  $\Lambda(\Gamma)$ . With this optimal stress test characterization, we then discuss some policy implications.

### 5.1 Optimal Credible Policy

For a given frequency  $N$ , the regulator always has an incentive to select the policy with the lowest possible strength  $k$ , provided that the policy remains credible  $k > \underline{k}(N)$ . Following the literature (see [Goldstein and Huang, 2016](#), for instance), we assume that the smallest feasible stress test strength that the regulator can choose, while still exceeding  $\underline{k}(N)$ , is given by  $\underline{k}(N)^+ = \underline{k}(N) + \varepsilon$  for some small  $\varepsilon > 0$ . In other words, there exists no feasible  $k$  within the interval  $(\underline{k}(N), \underline{k}(N)^+)$ . The benefit of conducting more tests comes from a lowered probability of failure in the worst equilibrium  $\mathcal{P}_f^W(N, k)$ . However, conducting more tests is costly. The optimal test balances the marginal cost and the benefit.

**Theorem 3.** *The socially optimal credible stress test policy is  $\Gamma^s = (N^s, \underline{k}(N^s)^+) \in \Gamma^C$ , in which*

$$N^s = \min \left\{ N \in \mathbb{Z}^+ : \underline{k}(N) - \underline{k}(N+1) \leq \frac{\Delta C(N)(\bar{\theta} - \underline{\theta})}{\pi(B + \chi)} \right\}. \quad (9)$$

The expected marginal benefit of one more test is given by the reduction in failure probability, multiplied by the sum of the bank's charter value  $B$  and the externality  $\chi$ , conditional on the occurrence of a liquidity distress event (with probability  $\pi$ ). Formally, this expected marginal benefit is:

$$\pi(B + \chi) \left( \mathcal{P}_f^W(N, \underline{k}(N)^+) - \mathcal{P}_f^W(N+1, \underline{k}(N+1)^+) \right) = \frac{\pi(B + \chi)}{\bar{\theta} - \underline{\theta}} (\underline{k}(N) - \underline{k}(N+1)).$$

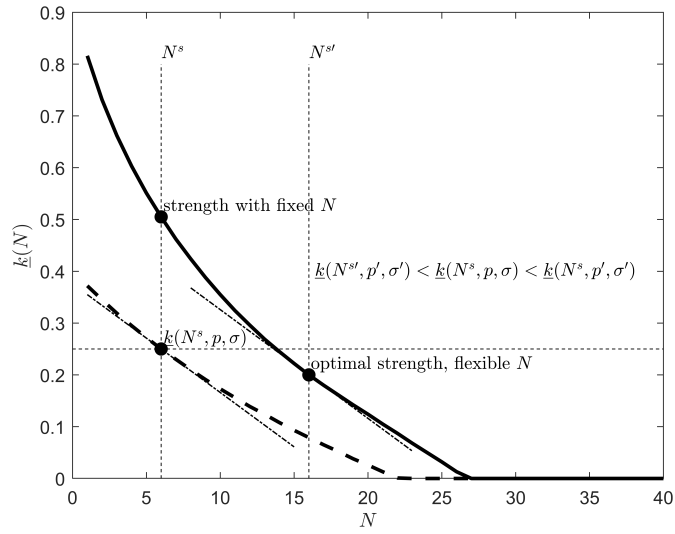
Since  $\underline{k}(N) - \underline{k}(N+1)$  is decreasing in test frequency  $N$  ([Theorem 2](#)), the expected marginal benefit declines as  $N$  increases. Moreover, because  $C(N)$  is convex, the marginal cost is increasing in  $N$ . As a result, there exists a unique optimal test frequency (9) that balances the marginal cost and benefit.

If the administrative cost of conducting more tests is small, the regulator may optimally choose  $N = \underline{N}(0)$ , and set the weakest possible test. This achieves the first best outcome since

no bank fails unless the fundamental warrants a failure ( $\theta < 0$ ). However, if the administrative costs are significant, the regulator might not conduct as many stress tests, and accordingly set a stringency above  $k = 0$ . Some banks will not pass such tests even though  $\theta > 0$ , and accordingly, the regulator does not achieve the first best outcome.

## 5.2 Policy Implication

The crucial policy implication of this paper is that it is not just the severity of the stress tests but also the ability to conduct such stress tests more often that helps the regulator in preventing runs. But why is frequency important? Below, in Figure 3, we present a numerical example to illustrate how accounting for frequency in our analysis can lead to markedly different policy implications. We consider a change in the market environment ( $p$  and  $\sigma$ ) such that under a given frequency of tests  $N$ , the regulator chooses a more stringent test than before, i.e.,  $\underline{k}(N)$  is higher than before. However, if the regulator is allowed to conduct more tests, she optimally chooses a higher frequency  $N' > N$  and a lower stringency than before, i.e.,  $\underline{k}(N')$  is lower than before.



**Figure 3. Optimal credible stress test policy when market conditions change.**

*Note:* Specification:  $f(x) = (2 + 4x) \cdot \mathbb{1}(-0.5 \leq x < 0) + (2 - 4x) \cdot \mathbb{1}(0 \leq x \leq 0.5)$ ,  $(B, \chi, \pi, \theta, \bar{\theta}) = (166, 71, 0.8, -1.5, 2.5)$ , and  $C(N) = N$ . The dashed thick line represents the minimum stringency when  $(p, \sigma) = (0.4, 0.05)$ , the solid thick line represents the minimum stringency when  $(p', \sigma') = (0.95, 1.41)$ . The tangency points represent the optimal stress test policy.

Even if the policymakers have the latitude to choose the frequency of stress tests, it is important to understand how they may be able to use this additional flexibility to enhance financial stability. Should the frequency of stress tests change depending on the macroeconomic environment, such as the likelihood of a distress event? What about bank-specific stress test

policies, depending on the bank’s systemic importance? The European Security and Market Authority (ESMA) guideline for stress test design suggests: “When deciding on the appropriate frequency, . . . the frequency should be adapted to the fund rather than a ‘one-size-fits-all’ approach being taken to all funds”<sup>20</sup> The following corollary shows how the likelihood of adverse shocks and the systemic importance of a bank affect the socially optimal stress test.

**Corollary 1.** *The regulator optimally chooses more frequent and less stringent tests (higher  $N^s$  and lower  $k(N^s)^+$ ) if (i) the shock is more likely to arrive (higher  $\pi$ ), and (ii) the bank failure can cause larger externalities to the banking system (higher  $\chi$ ).*

It directly follows from **Theorem 3** that higher  $\pi$  and  $\chi$  increases the expected marginal benefit from running more frequent tests, which gives the above result. Off-balance sheet exposures and a significant proportion of short-term wholesale funding may make a financial institution more susceptible to exogenous shocks, which we capture through the parameter  $\pi$ .<sup>21</sup> In an interconnected financial system, some institutions may impose a higher social cost on the system upon failure. We capture this externality through the parameter  $\chi$ . Because of such social costs, the central bank is more concerned about the failure of such interconnected financial institutions. Naturally, the gains from reducing the likelihood of failure are higher, and hence, despite the cost of additional tests, the central bank conducts more frequent credible tests which are less stringent.

Note that the prediction in **Corollary 1** considers banks that are exactly similar in all other aspects except their systemic importance and vulnerability to an external shock, i.e., their creditors have the same precision of private information ( $\sigma$ ) and the willingness to withdraw ( $p$ ). In practice, banks may differ in all of these aspects simultaneously, for example, creditors of systemically important banks may have more precise information, and therefore require more stringent testing. However, the central bank could increase the frequency and still conduct less stringent but credible tests. In **Figure C.1** in the Online Appendix **C**, we reproduce the current stress testing framework designed by the Federal Reserve. Bank holding companies that are systemically more important are categorized differently, monitored, and tested more frequently. By incorporating frequency in stress test design, our analysis offers a novel framework to assess the efficacy of stress test policy, i.e., whether it is sufficiently forceful and urgent in its supervision.

Finally, it is also important to highlight that without the regulator’s commitment, the stress test policy may not be effective. Both the United States and the European Union have enacted

<sup>20</sup>See bullet number 26 in the ESMA guidelines for stress testing document <https://www.esma.europa.eu/document/guidelines-liquidity-Stresstesting-in-ucits-and-aifs>

<sup>21</sup>In the guidelines for conducting stress tests for Alternative Investment Funds (AIFs), the European Security Market Authority (ESMA) acknowledges that a higher frequency of stress tests is called for when there is an emerging/imminent risk to fund liquidity.

laws mandating regular stress testing of financial institutions—such as the Dodd-Frank Act in the U.S. and the Single Supervisory Mechanism (SSM) in the EU. However, there were no such legal commitments during the initial implementation of stress tests. The relative success of the 2009 Supervisory Capital Assessment Program (SCAP) in the U.S., compared to the failure of the 2011 EU-wide stress tests in restoring market confidence, is often attributed to the U.S. commitment to future stress tests—already codified into law—whereas no such framework existed in Europe. As a result, market participants may have questioned the long-term solvency of European banks, despite their immediate success in the tests.<sup>22</sup>

## 6 Discussions and Extensions

### 6.1 Private Optimal Frequency

In practice, stress tests are sometimes delegated to banks themselves. They are granted discretion over key design elements such as the frequency and severity of the tests. In this section, we consider the bank optimally choosing the stress test policy instead of the regulator. Similar to the regulator, we assume that the bank anticipates that the agents will play the worst equilibrium, and she chooses a credible stress test policy. However, unlike the regulator, the bank does not internalize the social cost  $\chi$  associated with its failure. The bank's problem is to choose a credible stress policy to maximize

$$\Lambda_b(\Gamma) = B - \pi B \cdot \mathcal{P}_f^W(\Gamma) - C(N). \quad (10)$$

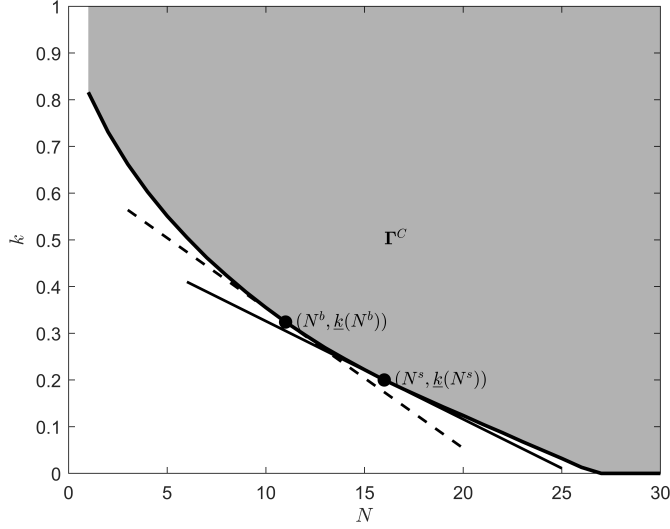
**Proposition 5.** *The bank's optimal choice of the stress test policy is  $(N^b, \underline{k}(N^b)^+)$ , in which*

$$N^b = \min \left\{ N \in \mathbb{Z}^+ : \underline{k}(N) - \underline{k}(N+1) \leq \frac{\Delta C(N)(\bar{\theta} - \underline{\theta})}{\pi B} \right\} \leq N^s. \quad (11)$$

Since the bank does not internalize the negative externality  $\chi$  associated with its failure—as the social planner does—and must also bear the cost of conducting stress tests, its privately optimal policy features a lower testing frequency  $N^b \leq N^s$ , and, correspondingly, a higher threshold for passing each test (i.e.,  $\underline{k}(N^b) \geq \underline{k}(N^s)$ ). **Figure 4** presents a numerical example that illustrates the gap between the bank's privately optimal design and the socially optimal policy choice.

Evidently, the greater the social cost of bank failure  $\chi$ , the more costly it becomes—from the social planner's perspective—to delegate the design of stress tests to the bank. This insight

<sup>22</sup>See the excerpt from [Abramovich \(2011\)](#): “One future test has been announced by the EBA for the first half of 2011. The United States, on the other hand, has enacted comprehensive legislation requiring stress testing of banks by regulators as well as self-testing. Although the United States legislation has not yet been implemented, the statute offers a regulatory framework for future stress testing. The United States has a legislative mandate with definite regularity in its application.”



**Figure 4. Socially optimal  $(N^s, \underline{k}(N^s))$  v.s. Private optimal  $(N^b, \underline{k}(N^b))$**

*Note:* Specification:  $f(x) = (2 + 4x) \cdot \mathbb{1}(-0.5 \leq x < 0) + (2 - 4x) \cdot \mathbb{1}(0 \leq x \leq 0.5)$ ,  $(B, \chi, \pi, \underline{\theta}, \bar{\theta}, p, \sigma) = (166, 71, 0.8, -1.5, 2.5, 0.95, 1.41)$ , and  $C(N) = N$ . The solid tangency lines represent the slope of the scaled cost function for the regulator and the dashed tangency lines represent the slope of the scaled cost function for the bank. The tangency points represent the optimal stress test policy.

provides a rationale for observed regulatory practices. For example, following the enactment of the “Economic Growth, Regulatory Relief, and Consumer Protection Act of 2018,” U.S. banks with less than \$100 billion in total consolidated assets are permitted to rely on internal, bank-led processes, whereas systemically important financial institutions (SIFIs)—including those with over \$250 billion in assets—remain subject to the Federal Reserve’s annual supervisory stress testing regime.<sup>23</sup>

## 6.2 Endogenous Asynchronous Debt Structure

Throughout this paper, we consider agents moving asynchronously at exogenously specified dates, as is the case under a staggered debt structure. In this section, we allow the bank to optimally choose the staggered debt structure. As we have seen in [Proposition 5](#), the bank wants less frequent tests compared to the social planner. Below, we argue that by endogenously choosing the debt structure, the bank can force the regulator to choose the privately optimal frequency. The game remains the same except at the beginning of the game, the bank chooses a staggered debt maturity structure  $M$ , under which a fraction  $1/M$  of debts mature at dates  $t = 0, \frac{1}{M}, \frac{2}{M}, \dots, \frac{M-1}{M}$ . When their debt matures, the agents decide whether to withdraw or rollover. After the bank chooses the debt structure  $M$ , the regulator chooses the stress test policy  $\Gamma = (N, k)$ , and then the agents play the dynamic coordination game. If no debts mature

<sup>23</sup>European Union’s “Capital Requirements Directive IV (CRD IV)” also use a similar rule. See [Figure C.1 \(Appendix B\)](#) for detail.

between two successive tests, then, the second test will not provide any additional information. Therefore, conducting more frequent tests will not reduce the likelihood of bank failure. This means a regulator cannot conduct more than  $M$  effective tests.

**Proposition 6.** *On the equilibrium path, the bank chooses the debt structure  $M^* = N^b \in [1, N^s]$ , and the regulator chooses the stress test policy  $\Gamma(N^b) = (N^b, \underline{k}(N^b)^+)$ , in which  $N^b$  is given in (11).*

The bank bears the cost of conducting  $N$  stress tests under the policy  $\Gamma = (N, k)$ . The equilibrium outcome is socially suboptimal because unlike the social planner the bank does not internalize the externality it imposes on the financial system (i.e.,  $\chi$ ). As a result, it selects a less staggered debt maturity structure than is socially optimal; specifically,  $M^* = N^b \leq N^s$ . This choice constrains the maximum feasible frequency of stress testing by the regulator. As shown in Proposition 6, the regulator's optimal policy is to match the test frequency with the bank's debt maturity structure. Notably, although the regulator sets the stress test frequency, the bank effectively implements its privately optimal frequency  $N^b$  through its initial maturity choice, consistent with the result in Proposition 5. In such cases, the regulator may turn to other policies (e.g., liquidity regulations) to complement the stress test policy and enhance social welfare.

### 6.3 Liquidity Regulation as a Complement

Next, we extend our theoretical framework by providing a microfoundation for the bank's fundamental  $\theta$ , and incorporating the effect of liquidity regulation on its distribution. Consider a bank with total liabilities of  $d$ , of which  $d_s$  matures in the short term (e.g., within 30 days). We normalize the potential outflow during a stress event to 1, and assume  $d_s < 1 < d$ . On the asset side, the bank can invest in a highly liquid asset (low yield), and a pledgeable asset (medium yield). Suppose the bank has  $1 + \eta$ , with  $\eta > 0$ , to allocate between the highly liquid and pledgeable assets. Under the LCR regulation, the bank must hold  $\lambda d_s$  in highly liquid assets, where  $\lambda$  is the regulatory ratio. The remaining  $1 + \eta - \lambda d_s$  is invested in the pledgeable asset.

The liquidity position is affected by a random shock  $\mu \sim \mathcal{U}[0, \bar{\mu}]$ , which directly reduces liquidity and lowers the pledgeability of the pledgeable asset to  $1 - \frac{\mu}{\bar{\mu}}$ . When the shock is most severe ( $\mu = \bar{\mu}$ ), the asset has zero pledgeability. Hence, the bank's post-shock liquidity position, or bank fundamental, is given by:

$$\theta = \underbrace{\lambda d_s}_{\text{Liquid asset}} - \underbrace{\mu}_{\text{Shock}} + \underbrace{\left(1 - \frac{\mu}{\bar{\mu}}\right)}_{\text{Pledgeability}} \cdot \underbrace{(1 + \eta - \lambda d_s)}_{\text{Pledgeable asset}}.$$

This implies that  $\theta \sim \mathcal{U}[\underline{\theta}, \bar{\theta}]$ , where  $\underline{\theta} = \lambda d_s - \bar{\mu}$  and  $\bar{\theta} = 1 + \eta$ . Thus, a higher LCR require-

ment (i.e., higher  $\lambda$ ) improves the bank's resilience to adverse shocks by increasing  $\underline{\theta}$ . While mandating very high values for the regulatory ratio  $\lambda$  could entail costs, we abstract from them here to emphasize how LCR complements stress testing by improving market confidence in the bank's fundamentals during distress.<sup>24</sup>

**Proposition 7.** *An increase in the liquidity coverage ratio (LCR),  $\lambda$ , leads to the following:*

1. *A higher socially optimal test frequency  $N^s$  and greater social welfare  $\Lambda(N^s, \underline{k}(N^s)^+)$  in the benchmark model without debt structure constraints; and*
2. *A higher optimal level of debt maturity asynchronicity  $M^* = N^b$  chosen by the bank, and greater social welfare  $\Lambda(N^b, \underline{k}(N^b)^+)$  in the extended model with debt structure constraints.*

**Proposition 7** shows that more stringent liquidity regulation can interact with stress test design to raise the socially optimal test frequency,  $N^s$ , thereby reducing the ex-ante probability of bank failure. In this sense, liquidity regulation complements the information disclosure role of frequent stress testing by facilitating creditor coordination, ultimately lowering the likelihood of coordination-driven bank runs and enhancing social welfare.

The argument behind this result is as follows. For any given stress test policy, a higher liquidity coverage ratio (LCR) directly improves welfare by reducing the probability of bank failure, owing to its positive effect on the threshold  $\underline{\theta}$  (see (4)). In addition, an increase in  $\underline{\theta}$  implies more optimistic prior belief; the marginal benefit of raising the test frequency increases, since more frequent stress tests become more effective at lowering the ex-ante likelihood of failure. As a result, the social planner chooses a higher frequency  $N^s$ .<sup>25</sup> Although a higher frequency entails greater administrative costs, the envelope condition implies that the social welfare increases.<sup>26</sup>

A similar logic applies when the bank endogenously chooses the debt asynchronicity level,  $M^* = N^b$ . However, in this case, the envelope condition no longer applies, as  $M^* = N^b$  is chosen to maximize the bank's private objective function  $\Lambda_b(M, (M, \underline{k}(M)^+))$  (see **Proposition 6**), rather than to maximize social welfare. Nevertheless, a higher LCR implies that there are more frequent tests  $N^b$ , which leads to greater marginal social benefit relative to the cost of stress tests. In summary, a higher LCR induces the bank to adopt a more asynchronous debt structure, thereby easing the regulator's constraints and indirectly improving social welfare.

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<sup>24</sup>We require that  $\bar{\mu} > d_s + \sigma$  so that the lower bound of the bank's fundamental  $\underline{\theta} < -\sigma$

<sup>25</sup>Technically, a higher LCR raises  $\underline{\theta}$ , which in turn increases the socially optimal test frequency  $N^s$ , as the upper bound on  $\underline{k}(N) - \underline{k}(N+1)$  declines (see (9)).

<sup>26</sup>We ignore the cost of liquid reserves to make the insight stark. If there is some cost of liquidity then the regulator will choose an optimal liquidity coverage ratio. However, in practice, the social cost of systemic failure is much greater than the cost of liquidity. Therefore, it is reasonable to conclude that the social welfare increases with the liquidity coverage ratio.

## 7 Conclusion

This paper studies stress testing as an adversarial dynamic information design problem in a bank-run environment with incomplete information and asynchronous moves. It introduces frequency as a central—and previously underexplored—policy lever in the design of credible stress tests. In our framework, a regulator conducts repeated pass/fail tests during a distress period, and credibility demands that no bank passing a test subsequently fails, even under the worst equilibrium. We show that frequent testing disciplines worst-case beliefs: by limiting how much damage any skeptical cohort can inflict before the next checkpoint, higher frequency reduces the minimum stringency required for credibility. This result bridges two existing benchmarks—static persuasion with a single test and the limiting case of arbitrarily frequent disclosure—by characterizing the full trade-off between frequency and stringency. The practical implication is that credibility need not be achieved solely through harsher stress scenarios. Regulators can substitute frequency for stringency, allowing more banks to pass while still maintaining credibility, thereby lowering the ex-ante probability of inefficient, run-driven failures. Because testing is costly, the optimal policy features a finite frequency that balances diminishing credibility gains against administrative costs, and this frequency increases with systemic risk and bank fragility.

Understanding the design of stress tests is essential due to their widespread use in practice; however, it is also important to acknowledge their limitations. Our regulator implements a credible stress test policy by conducting uniform assessments at regular intervals and publicly disclosing pass or fail outcomes to stakeholders. The passing criteria for these tests are monotonic, meaning that if a weaker bank passes, a stronger bank will also pass. While these features are practical in the context of bank runs, they may not be entirely without loss of generality. In the context of a static coordination game, [Inostroza and Pavan \(2025\)](#) demonstrates that non-monotonic disclosure policies can outperform monotonic disclosure policies such as stress tests. Furthermore, [Li, Song and Zhao \(2023\)](#) finds that an optimal policy in a coordination game may involve disclosing different information to different agents. Whether non-credible or non-uniform disclosure can outperform uniform credible disclosure remains an open question.

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## Appendix

*Proof of Lemma 1.* First, we show that under any given monotone strategy profile  $\hat{\mathbf{s}} = \{\hat{s}_n(\zeta^n)\}_{n=1}^N$ , a higher fundamental leads to more favorable information. Specifically, for  $\theta_2 \geq \theta_1$ , it follows that  $\zeta^n(\theta_2; \hat{\mathbf{s}}) \succeq \zeta^n(\theta_1; \hat{\mathbf{s}})$  for all  $n \in \{1, 2, \dots, N\}$ .

This result can be derived by induction. Since  $\theta_2 \geq \theta_1$  implies  $\zeta^1(\theta_2; \hat{\mathbf{s}}) \succeq \zeta^1(\theta_1; \hat{\mathbf{s}})$ , it suffices to show that if  $\zeta^m(\theta_2; \hat{\mathbf{s}}) \succeq \zeta^m(\theta_1; \hat{\mathbf{s}})$  holds for all  $m \leq n$ , then it also holds for  $m = n + 1$ . By the definition of monotone strategies,  $\zeta^m(\theta_2; \hat{\mathbf{s}}) \succeq \zeta^m(\theta_1; \hat{\mathbf{s}})$  for all  $m \leq n$  implies that  $\hat{s}_m(\zeta^m(\theta_2; \hat{\mathbf{s}})) \leq \hat{s}_m(\zeta^m(\theta_1; \hat{\mathbf{s}}))$  for all  $m \leq n$ . Consequently, we obtain

$$\frac{1}{N}F\left(\frac{\hat{s}_m(\zeta^m(\theta_2; \hat{\mathbf{s}})) - \theta_2}{\sigma}\right) \leq \frac{1}{N}F\left(\frac{\hat{s}_m(\zeta^m(\theta_1; \hat{\mathbf{s}})) - \theta_1}{\sigma}\right), \text{ for all } m = 1, \dots, n.$$

In other words, the mass of attacks in group  $m$  ( $m = 1, \dots, n$ ) under  $\theta_2$  is weakly lower than that under  $\theta_1$ . As a result,

$$\begin{aligned} \theta^{\frac{n}{N}}(\theta_2; \hat{\mathbf{s}}) &= \theta_2 - \frac{1}{N} \sum_{m=1}^n F\left(\frac{\hat{s}_m(\zeta^m(\theta_2; \hat{\mathbf{s}})) - \theta_2}{\sigma}\right) \\ &\geq \theta_1 - \frac{1}{N} \sum_{m=1}^n F\left(\frac{\hat{s}_m(\zeta^m(\theta_1; \hat{\mathbf{s}})) - \theta_1}{\sigma}\right) = \theta^{\frac{n}{N}}(\theta_1; \hat{\mathbf{s}}), \end{aligned} \quad (\text{A.1})$$

Therefore, we conclude that  $\zeta_{n+1}(\theta_2; \hat{\mathbf{s}}) \succeq \zeta_{n+1}(\theta_1; \hat{\mathbf{s}})$ .

Next, we prove that for any monotone strategy profile  $\hat{\mathbf{s}}$ , there exists a set of history-independent fundamental thresholds  $\{\underline{\theta}_n(\hat{\mathbf{s}})\}_{n=0}^N$  such that for  $n \in \{0, \dots, N-1\}$ ,  $\zeta_{n+1} = P$  if and only if  $\theta \geq \underline{\theta}_n$ , and  $\theta^1 \geq 0$  if and only if  $\theta \geq \underline{\theta}_N \equiv \hat{\theta}$ .

Since the bank passes the  $(n+1)$ th stress test if and only if  $\theta^{\frac{n}{N}}(\theta; \hat{\mathbf{s}}) \geq (1 - \frac{n}{N})k$  for  $n = 0, \dots, N-1$ , and survives if and only if  $\theta^1(\theta; \hat{\mathbf{s}}) \geq 0$ , it is sufficient to show that  $\theta^{\frac{n}{N}}(\theta; \hat{\mathbf{s}})$  is increasing in  $\theta$  for all  $n = 0, 1, \dots, N$ . It follows directly that  $\theta^0(\theta; \hat{\mathbf{s}}) = \theta$  and it is increasing in  $\theta$ . Note that, for  $n = 1, \dots, N$ , given  $\theta$  and strategy profile  $\hat{\mathbf{s}}$  the inequality (A.1) implies that  $\theta^{\frac{n}{N}}(\theta; \hat{\mathbf{s}})$  is an increasing function of  $\theta$ . As  $\theta^{\frac{n}{N}}(\theta; \hat{\mathbf{s}})$  is increasing in  $\theta$ , and we can express  $\underline{\theta}_n(\hat{\mathbf{s}})$  as

$$\underline{\theta}_n(\hat{\mathbf{s}}) = \inf \left\{ \theta \in [\underline{\theta}, \bar{\theta}] \mid \theta^{\frac{n}{N}}(\theta; \hat{\mathbf{s}}) \geq \left(1 - \frac{n}{N}\right)k \right\}, \quad n = 0, 1, \dots, N, \quad (\text{A.2})$$

such that  $\zeta_{n+1} = P$  if and only if  $\theta \geq \underline{\theta}_n(\hat{\mathbf{s}})$  and  $\theta^1 \geq 0$  if and only if  $\theta \geq \underline{\theta}_N(\hat{\mathbf{s}})$ . ■

*Proof of Lemma 2.* Suppose the other stakeholders attack if and only if the bank has failed any previous test, regardless of their private signals. In other words, they follow a monotone strategy  $\hat{s}_{-i} \equiv \{\hat{s}_{n,j}(\zeta^n)\}_{j \neq i, n=1, \dots, N}$  defined as follows:

$$\hat{s}_{n,j}(\zeta^n) = \begin{cases} k - \frac{\sigma}{2}, & \text{if } \zeta_1 = \dots = \zeta_n = P \\ 1 + \frac{\sigma}{2}, & \text{otherwise} \end{cases}, \text{ for all } j \neq i \text{ and } n = 1, 2, \dots, N.$$

For a bank with fundamental  $\theta \geq k$ , it passes the first test, so the mass of attackers in the first group is 0. As a result, the residual fundamental at  $t = \frac{1}{N}$  is  $\theta^{\frac{1}{N}}(\theta; \hat{s}_{-i}) = \theta$ , which satisfies  $\theta > (1 - \frac{1}{N})k$ , allowing it to pass the second test. This process continues, ensuring that any bank with  $\theta \geq k$  passes all tests and survives.

Conversely, a bank with fundamental  $\theta < k$  fails the first test. With all stakeholders attacking (except for stakeholder  $i$ , whose influence is negligible), the residual fundamental at  $t = \frac{1}{N}$  is  $\theta^{\frac{1}{N}}(\theta; \hat{s}_{-i}) = \theta - \frac{1}{N}$ , which is less than  $(1 - \frac{1}{N})k$ . Consequently, it fails the second test and continues to fail all subsequent tests, ultimately leading to failure.

Thus, given that other stakeholders follow the monotone strategy  $\hat{s}_{-i}$ , the bank survives if and only if  $\theta \geq k$ , implying:

$$k = \underline{\theta}_0 = \underline{\theta}_1 = \dots = \underline{\theta}_N = \hat{\theta}.$$

Therefore, the best response for stakeholder  $i$  is to adopt the same strategy, i.e.,  $\hat{s}_{n,i}(\zeta^n) = \hat{s}_{n,j}(\zeta^n)$  for all  $n$  and  $\zeta^n$ .

For the “only if” argument, consider any policy  $\Gamma = (N, k)$ . If the equilibrium survival threshold is  $\hat{\theta}(e(\Gamma)) = k$ , then, agents must play the obedient equilibrium. To see this, observe that when  $\hat{\theta}(e(\Gamma)) = k$ , in equilibrium  $e(\Gamma)$ , all agents will always attack (never attack) if the bank fails (passes) the first test since that test result implies that  $\theta < \hat{\theta} = k$  ( $\theta \geq \hat{\theta} = k$ ), regardless of their private signals. Hence, the bank passes the second test if and only if  $\theta \geq \hat{\theta} = k$ . This implies that, similar to agents’ response to the first test result, the agents in equilibrium must follow the test results in the second and subsequent tests. Therefore,  $\hat{\theta}(e(\Gamma)) = k$  implies that agents play the obedient equilibrium,  $e(\Gamma) = e^o(\Gamma)$ . ■

*Proof of Proposition 1.* If  $\mathcal{E}^D(\Gamma) \neq \emptyset$ , then the worst equilibrium features  $\hat{\theta}(e^W(\Gamma)) > k$ . By definition of credible stress tests, such a policy  $\Gamma$  is not credible because there exists  $\theta \in (k, \hat{\theta}(e^W(\Gamma)))$  such that a bank with this post-shock fundamental would pass the first test (i.e.,  $\theta \geq k$ ) but ultimately fail (i.e.,  $\theta < \hat{\theta}(e^W(\Gamma))$ ). Conversely, if  $\mathcal{E}^D(\Gamma) = \emptyset$ , then the worst equilibrium under this policy is the obedient equilibrium, i.e.,  $e^W(\Gamma) = e^o(\Gamma)$ . Under this equilibrium, the bank passes all tests and survives the distress period if and only if  $\theta \geq k$ , implying that the stress test policy is credible. ■

*Proof of Lemma 3.* The proof consists of four steps.

**Step 1:** We show that in a monotone equilibrium, agents must attack less (more) aggressively when they see an additional pass (fail) stress test result. We will prove it for additional pass result. An analogous argument can be made for a fail result. Suppose the history of stress test result is  $\zeta^{n-1}$ , and this reveals  $\theta \in \Theta(\zeta^{n-1})$  and the private signals agents might receive for such  $\theta$  are  $s \in \mathbb{S}(\zeta^{n-1})$ .

Suppose, for contradiction, under this equilibrium  $\hat{s}$ , agents attack more aggressively when they see an additional pass stress test result, i.e.,

$$F\left(\frac{\hat{s}_n(\zeta^{n-1} \cup \{P\}) - \theta}{\sigma}\right) > F\left(\frac{\hat{s}_{n-1}(\zeta^{n-1}) - \theta}{\sigma}\right), \text{ or, } \hat{s}_n(\zeta^{n-1} \cup \{P\}) > \hat{s}_{n-1}(\zeta^{n-1}).$$

Case 1:  $F\left(\frac{\hat{s}_{n-1}(\zeta^{n-1}) - \theta}{\sigma}\right) > 0$  for some  $\theta \in \Theta(\zeta^{n-1})$ , i.e., there is positive attack from group  $(n-1)$  after observing  $\zeta^{n-1}$ . Then, in equilibrium, the marginal agent must be indifferent between attacking and not attacking. This means the marginal agent believes the bank will survive with probability  $p$ . By Lemma 1, the bank survives iff  $\theta \geq \hat{\theta}$ . Therefore, we must have

$$\mathbb{P}(\theta \geq \hat{\theta} | \hat{s}_{n-1}, \zeta^{n-1}) = p.$$

If  $\hat{s}_n(\zeta^{n-1} \cup \{P\}) > \hat{s}_{n-1}(\zeta^{n-1})$ , then it follows from Assumption 1 that regardless of the information revealed by the stress test results  $\zeta^{n-1}$ ,

$$\mathbb{P}(\theta \geq \hat{\theta} | \hat{s}_n, \zeta^{n-1}) > \mathbb{P}(\theta \geq \hat{\theta} | \hat{s}_{n-1}, \zeta^{n-1}) = p.$$

This means if group  $n$  learns nothing new from the  $n$ -th test, the marginal agent  $\hat{s}_n(\zeta^{n-1} \cup \{P\})$  strictly prefers not attacking. We know from Lemma 1 that when the bank passes the  $n$ -th test, the agents learn  $\theta \geq \underline{\theta}_{n-1}$ , which makes them weakly more optimistic. Therefore,

$$\mathbb{P}(\theta \geq \hat{\theta} | \hat{s}_n, \zeta^{n-1}, \theta \geq \underline{\theta}_{n-1}) \geq \mathbb{P}(\theta \geq \hat{\theta} | \hat{s}_n, \zeta^{n-1}) > p.$$

This means no agent in group  $n$  will attack after observing  $\zeta^{n-1} \cup \{P\}$ , i.e.,  $F\left(\frac{\hat{s}_n(\zeta^{n-1} \cup \{P\}) - \theta}{\sigma}\right) = 0$ , hence contradiction.

Case 2:  $F\left(\frac{\hat{s}_{n-1}(\zeta^{n-1}) - \theta}{\sigma}\right) = 0$  for all  $\theta \in \Theta(\zeta^{n-1})$ , i.e., there is zero attack from group  $(n-1)$  after observing  $\zeta^{n-1}$ . This means following  $\zeta^{n-1}$ , the bank will always pass the  $n$ -th test. In other words, passing the  $n$ -th test reveals no new information, i.e.,  $\Theta(\zeta^{n-1} \cup \{P\}) = \Theta(\zeta^{n-1})$  and the private signals agents might receive for such  $\theta$  are  $\mathbb{S}(\zeta^{n-1} \cup \{P\}) = \mathbb{S}(\zeta^{n-1})$ . In this case  $\hat{s}_{n-1}(\zeta^{n-1}) = \inf\{\mathbb{S}(\zeta^{n-1})\}$  and any agent with private signal  $s > \hat{s}_{n-1}(\zeta^{n-1})$  strictly prefers not attacking, i.e.,  $\mathbb{P}(\theta \geq \hat{\theta} | s, \zeta^{n-1}) > p$ . Since agents learn nothing new from the  $n$ -th pass,  $\mathbb{P}(\theta \geq \hat{\theta} | s, \zeta^{n-1} \cup \{P\}) > p$  for all  $s > \inf\{\mathbb{S}(\zeta^{n-1} \cup \{P\})\}$ . Therefore, there will

be no attack from group  $n$  after observing  $\zeta^{n-1} \cup \{P\}$ , i.e,  $F\left(\frac{\hat{s}_n(\zeta^{n-1} \cup \{P\}) - \theta}{\sigma}\right) = 0$  for all  $\theta \in \Theta(\zeta^{n-1} \cup \{P\})$ , hence contradiction.

**Step 2:** It follows from Step 1 that

$$k = \underline{\theta}_0 \leq \underline{\theta}_1 \leq \dots \leq \underline{\theta}_{N-1} \leq \underline{\theta}_N = \hat{\theta}.$$

Suppose, for contradiction, this is not true. Then, there is a cutoff equilibrium with  $\hat{\theta} > k$ , in which for some  $n$ ,  $\underline{\theta}_{n-1} > \underline{\theta}_n$ .

Let  $n$  be the first test for which this inequality holds.  $n = 1$  is not possible. If  $\underline{\theta}_1 < \underline{\theta}_0 = k$ , then for  $\theta \in (\underline{\theta}_1, k)$  the bank fails the first test but passes the second test, which means less than  $k$  proportion of agents attack from the first group. However, since  $\hat{\theta} > k$ , failing the first test means the bank cannot survive in the end. Therefore, all agents in group 1 must have attacked, giving us a contradiction. Therefore, we have  $\underline{\theta}_1 \geq \underline{\theta}_0 = k$ .

Consider  $n \geq 2$  and  $\theta \in (\max\{\underline{\theta}_n, \underline{\theta}_{n-2}\}, \underline{\theta}_{n-1})$ . When the fundamental is  $\theta$ , in equilibrium, the bank will pass the first  $(n-1)$  tests, fail the  $n$ -th test, and pass the  $(n+1)$ th test. This means under fundamental  $\theta$ , the history of stress tests will be  $\zeta^{n'}(\theta, \hat{s}) = \{P, \dots, P\}$  for all  $n' \leq n-1$ ,  $\zeta^n = \{P, \dots, P, F\}$ , and therefore,

$$\theta - \frac{1}{N} \sum_{n'=1}^{n-2} F\left(\frac{\hat{s}_{n'}(\zeta^{n'}) - \theta}{\sigma}\right) \geq k \left(1 - \frac{n-2}{N}\right) \quad (\text{A.3})$$

$$\theta - \frac{1}{N} \sum_{n'=1}^{n-1} F\left(\frac{\hat{s}_{n'}(\zeta^{n'}) - \theta}{\sigma}\right) < k \left(1 - \frac{n-1}{N}\right) \quad (\text{A.4})$$

$$\theta - \frac{1}{N} \sum_{n'=1}^n F\left(\frac{\hat{s}_{n'}(\zeta^{n'}) - \theta}{\sigma}\right) \geq k \left(1 - \frac{n}{N}\right) \quad (\text{A.5})$$

(A.3) – (A.4) gives us

$$F\left(\frac{\hat{s}_{n-1}(\zeta^{n-1} = \{P, \dots, P\}) - \theta}{\sigma}\right) > k$$

and (A.4) – (A.5) gives us

$$F\left(\frac{\hat{s}_n(\zeta^n = \{P, \dots, P, F\}) - \theta}{\sigma}\right) < k.$$

The first inequality means the proportion of attack from the  $(n-1)$ th group must exceed  $k$  whereas the second inequality means the proportion of attack from the  $n$ -th group must be below  $k$ . These two inequalities can hold together only if  $\hat{s}_n(\zeta^n = \{P, \dots, P, F\}) < \hat{s}_{n-1}(\zeta^{n-1} = \{P, \dots, P\})$ . Since from step 1 we know that in a monotone equilibrium, agents in later group attacks more aggressively after seeing additional unfavorable stress test result, the above in-

equality cannot hold, giving us a contradiction. Therefore,  $\underline{\theta}_n$  must be weakly increasing for all  $n$ .

**Step 3:** Since in a disobedient equilibrium,  $\underline{\theta}_{n-1} \leq \hat{\theta}$  for any  $n$ , a bank cannot survive unless it pass all the tests. Therefore, if a bank fails the  $n$ -th test, then any agent that observes this stress test result understands that the bank cannot survive, and hence, always attacks. Accordingly, in a disobedient equilibrium,  $\hat{s}_n(\zeta^n) = \underline{\theta}_{n-1} + \frac{\sigma}{2}$  for any  $\zeta^n \neq \{P\}^n$ . If the bank has passed all the tests so far, i.e.,  $\zeta^n = \{P\}^n$ , an agent may still attack if his private signal is not sufficiently high, i.e.,  $s_i < \hat{s}_n(\zeta^n = \{P\}^n)$ . Since the agent always attack whenever the bank fails a previous test, the equilibrium strategy can be succinctly represented by  $N$  information thresholds

$$\hat{\mathbf{s}} = (\hat{s}_n)_{n=1}^N,$$

where  $\hat{s}_n$  corresponds to the history in which the bank has passed all previous tests; that is,  $\hat{s}_n \equiv \hat{s}_n(\zeta^n = \{P\}^n)$ .

**Step 4:** Since in any disobedient equilibrium,  $\underline{\theta}_n$  is non-decreasing, a bank cannot pass the  $n$ -th test unless it has passed all the  $n' < n$  tests. If  $\underline{\theta}_{n-1} < \underline{\theta}_n$ , then for  $\theta \in (\underline{\theta}_{n-1}, \underline{\theta}_n)$ , the bank passes the first  $n$  tests but fails the  $(n+1)$ th test. On the other hand, if  $\underline{\theta}_{n-1} = \underline{\theta}_n$  then whenever the bank passes the first  $n$  tests it also passes the  $(n+1)$ th test. Consider the lowest  $n$  such that  $\underline{\theta}_{n-1} = \underline{\theta}_n$ . Recall that passing the first  $n$  tests means  $\theta$  is such that the bank can withstand  $k$  fraction of attack from the agents that are left, i.e.,

$$\theta - \frac{1}{N} \sum_{n'=1}^{n-1} F\left(\frac{\hat{s}_{n'} - \theta}{\sigma}\right) \geq k \left(1 - \frac{n-1}{N}\right).$$

Since the LHS is increasing in  $\theta$ , the survival threshold  $\underline{\theta}_{n-1}$  is such that the above holds with equality, i.e.,

$$\underline{\theta}_{n-1} - \frac{1}{N} \sum_{n'=1}^{n-1} F\left(\frac{\hat{s}_{n'} - \underline{\theta}_{n-1}}{\sigma}\right) = k \left(1 - \frac{n-1}{N}\right). \quad (\text{A.6})$$

Since under  $\underline{\theta}_{n-1} = \underline{\theta}_n$  passing the first  $n$  tests guarantees the bank passes the  $(n+1)$ th test, when  $\theta = \underline{\theta}_{n-1}$ , the attack from group  $n$  must be such that

$$\underline{\theta}_{n-1} - \frac{1}{N} \sum_{n'=1}^{n-1} F\left(\frac{\hat{s}_{n'} - \underline{\theta}_{n-1}}{\sigma}\right) - \frac{1}{N} F\left(\frac{\hat{s}_n - \underline{\theta}_{n-1}}{\sigma}\right) \geq k \left(1 - \frac{n}{N}\right). \quad (\text{A.7})$$

(A.6) — (A.7) gives

$$F\left(\frac{\hat{s}_n - \underline{\theta}_{n-1}}{\sigma}\right) \leq k.$$

Therefore, if in equilibrium,  $\underline{\theta}_{n-1} = \underline{\theta}_n$ , then when  $\theta = \underline{\theta}_{n-1}$ , the fraction of attack from group  $n$  is less than the fraction of attack the bank can sustain from the rest of the agents as revealed

by the stress test. Since from step 1, we know that  $\hat{s}_{n+1} \leq \hat{s}_n$ , the fraction of attack from group  $(n+1)$  will be even smaller, i.e.,

$$F\left(\frac{\hat{s}_{n+1} - \underline{\theta}_{n-1}}{\sigma}\right) \leq F\left(\frac{\hat{s}_n - \underline{\theta}_{n-1}}{\sigma}\right) \leq k.$$

Accordingly, if in equilibrium,  $\underline{\theta}_{n-1} = \underline{\theta}_n$ , then

$$\underline{\theta}_{n-1} - \frac{1}{N} \sum_{n'=1}^{n+1} F\left(\frac{\hat{s}_{n'} - \underline{\theta}_{n-1}}{\sigma}\right) \geq k \left(1 - \frac{n+1}{N}\right). \quad (\text{A.8})$$

This means when the bank passes the first  $n$  tests, it also passes the  $(n+1)$ th test and the  $(n+2)$ th test, i.e.,  $\underline{\theta}_{n-1} = \underline{\theta}_n = \underline{\theta}_{n+1}$ . Continuing the same argument, one can establish that once the bank passes the first  $n$  tests, it will pass all the subsequent tests and survive, i.e.,  $\underline{\theta}_{n-1} = \underline{\theta}_n = \underline{\theta}_{n+1} = \dots = \underline{\theta}_N = \hat{\theta}$ . This means once the bank passes the first  $n$  tests, agents learn that the bank will survive, and therefore, no agent starting from group  $n$  will attack once they see the bank has passed all the tests so far. We call these groups obedient. The above argument establishes that in equilibrium if one group is obedient, then it must be that all the subsequent groups are also obedient. In other words, there exists a last disobedient group  $\bar{n} \in \{1, 2, \dots, N\}$  such that  $k = \underline{\theta}_0 < \underline{\theta}_1 < \dots < \underline{\theta}_{\bar{n}} = \underline{\theta}_{\bar{n}+1} = \dots = \underline{\theta}_N = \hat{\theta}$ . ■

*Proof of Proposition 2.* Under a stress test policy  $\Gamma = (N, k)$ , if a disobedient equilibrium  $e \in \mathcal{E}^D(\Gamma)$  exists, then it follows from Lemma 3 that the disobedient equilibrium consists of  $N$  thresholds  $\hat{s} = \{\hat{s}_n\}_{n=1}^N$  and a corresponding last disobedient group  $\bar{n}(\hat{s}) \in \{1, 2, \dots, N\}$  such that

$$k = \underline{\theta}_0 < \underline{\theta}_1 < \dots < \underline{\theta}_{\bar{n}} = \underline{\theta}_{\bar{n}+1} = \dots = \underline{\theta}_N = \hat{\theta}.$$

Consider any disobedient group  $n \leq \bar{n}$ . The bank passes the first  $n$  tests iff  $\theta \geq \underline{\theta}_{n-1}$ , where  $\underline{\theta}_{n-1}$  is defined in (A.6). In equilibrium, an agent  $i$  in group  $n$  wants to attack iff  $s_i < \hat{s}_n$ . Notice that she believes that the bank will survive with probability

$$P(\theta \geq \hat{\theta} | s_i, \theta \geq \underline{\theta}_{n-1}) = \frac{F\left(\frac{s_i - \hat{\theta}}{\sigma}\right)}{F\left(\frac{s_i - \underline{\theta}_{n-1}}{\sigma}\right)}.$$

It follows from log-concavity that this belief is increasing in  $s_i$ . The lowest signal an agent  $i$  in group  $n$  may receive is  $s_i = \underline{\theta}_{n-1} - \sigma/2$ . Since group  $n$  is disobedient  $\hat{s}_n > \underline{\theta}_{n-1} - \sigma/2$ , and the marginal agent  $\hat{s}_n$  must believe that the probability of survival

$$\frac{F\left(\frac{\hat{s}_n - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_n - \underline{\theta}_{n-1}}{\sigma}\right)} = p.$$

This makes the marginal agent indifferent between attacking and not attacking. While all the agents with  $s_i > \hat{s}_n$  prefer not attacking and all the agents with  $s_i < \hat{s}_n$  prefer attacking. For any obedient group  $n \geq \bar{n} + 1$  (if there is one), the cutoff signal is  $\hat{s}_n = \underline{\theta}_{n-1} - \sigma/2$ . ■

*Proof of Theorem 1.* The proof follows four steps.

**Step 1:** Consider the disobedient equilibrium with  $\bar{n} = 1$ , i.e., the first group is the last disobedient group. Under this equilibrium, if the bank can survive the attack from group 1 and pass the second test, it will survive, i.e.,  $\underline{\theta}_1 = \hat{\theta}$ . Therefore the survival threshold  $\hat{\theta}$  is such that (see ( $F_n$ ))

$$\hat{\theta} = \frac{x}{N} + k \left(1 - \frac{1}{N}\right), \text{ where } x = F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right)$$

is the fraction of attack from group 1 when  $\theta = \hat{\theta}$ . In a disobedient equilibrium  $\hat{\theta} > k$ , which means fraction of attack at the survival threshold  $x > k$ . Consider the marginal agent  $\hat{s}_1$  in group 1. When the bank passes the first test, he learns  $\theta \geq k$ . However, this is not sufficient for survival. If  $\theta \geq \underline{\theta}_1 = \hat{\theta}$ , then the bank can withstand the attack from group 1, and pass the second test, and then, survive. Therefore, he believes that the bank will survive with probability

$$\mathbb{P}(\theta \geq \hat{\theta} \mid \hat{s}_1, \theta \geq k) = \frac{F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_1 - k}{\sigma}\right)} = \frac{F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma} + \frac{\hat{\theta} - k}{\sigma}\right)} = \frac{x}{F\left(F^{-1}(x) + \frac{x-k}{N\sigma}\right)} = G(x, N, k).$$

If there exists a disobedient equilibrium, in which the first group is the last disobedient group ( $\bar{n} = 1$ ), then there must exist  $x \in (k, 1]$  such that the marginal agent in the first group is indifferent (see ( $I_n$ )), i.e.,

$$G(x, N, k) = p.$$

**Step 2:** Consider a disobedient equilibrium in which the last disobedient group  $\bar{n} > 1$ . After seeing that the bank has passed the first  $\bar{n}$  tests, the agents learn  $\theta \geq \underline{\theta}_{\bar{n}-1}$ . By definition, when  $\theta = \underline{\theta}_{\bar{n}-1}$ , the residual fundamental is just enough to withstand  $k$  fraction of attack from the agents who are yet to make their decisions, i.e.,

$$\theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}-1}) = \left(1 - \frac{\bar{n}-1}{N}\right)k.$$

However, this is not enough for survival. If the bank can withstand the attack from group  $\bar{n}$  and pass the next test, then it survives, i.e.,  $\underline{\theta}_{\bar{n}} = \hat{\theta}$ . By definition, when  $\theta = \underline{\theta}_{\bar{n}}$ , the residual fundamental is just enough to sustain the attack from the  $\bar{n}$ -th group and then pass the next

test if there is one (see ( $F_n$ )); that is ,

$$\theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}}) = \frac{x}{N} + \left(1 - \frac{\bar{n}}{N}\right)k, \text{ where } x = F\left(\frac{\hat{s}_{\bar{n}} - \underline{\theta}_{\bar{n}}}{\sigma}\right)$$

is the fraction of attack from the last disobedient group  $\bar{n}$  when the fundamental is at the survival threshold, i.e.,  $\theta = \underline{\theta}_{\bar{n}} = \hat{\theta}$ . Therefore, the distance between the two residual fundamentals is

$$\theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}}) - \theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}-1}) = \frac{x-k}{N}.$$

When  $\theta$  is higher agents are less likely to observe lower signals, and therefore, in a monotone equilibrium, aggregate attack is lower.<sup>27</sup> This leads to the widening gap property (**W**). It then follows from (**W**) that

$$\underline{\theta}_{\bar{n}} - \underline{\theta}_{\bar{n}-1} \leq \theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}}) - \theta^{\frac{\bar{n}-1}{N}}(\theta = \underline{\theta}_{\bar{n}-1}) = \frac{x-k}{N}. \quad (\text{A.9})$$

When  $\bar{n} = 1$ , the distance between the threshold to survive and the threshold to pass the first test is  $\frac{x-k}{N}$ . That is, the above holds with equality. When  $\bar{n} > 1$ , for any fraction of aggregate attack from the last disobedient group  $x \in (k, 1]$ , the gap between the threshold for survival  $\underline{\theta}_{\bar{n}}$  and the threshold for passing all the tests so far  $\underline{\theta}_{\bar{n}-1}$  is lower than the case with  $\bar{n} = 1$ . It follows from (A.9) that the marginal agent in the last disobedient group more optimistic about the bank's survival, i.e.,

$$\mathbb{P}(\theta \geq \hat{\theta} \mid \hat{s}_{\bar{n}}, \theta \geq \underline{\theta}_{\bar{n}-1}) = \frac{F\left(\frac{\hat{s}_{\bar{n}} - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_{\bar{n}} - \underline{\theta}_{\bar{n}-1}}{\sigma}\right)} = \frac{F\left(\frac{\hat{s}_{\bar{n}} - \hat{\theta}}{\sigma}\right)}{F\left(\frac{\hat{s}_{\bar{n}} - \hat{\theta}}{\sigma} + \frac{\hat{\theta} - \underline{\theta}_{\bar{n}-1}}{\sigma}\right)} \geq \frac{x}{F\left(F^{-1}(x) + \frac{x-k}{N\sigma}\right)} = G(x, N, k).$$

Thus,  $G(x, N, k)$  serves as a lower bound for the marginal agent's belief in the last disobedient group ( $\bar{n}$ ) that the bank will survive.

**Step 3:** Suppose that **Condition C** is violated. Then there exists  $x \in (k, 1]$  such that  $G(x, N, k) \leq p$ . Since  $G(k, N, k) = G(1, N, k) = 1$ , by continuity, there exists  $x^* \in (k, 1]$  for which  $G(x^*, N, k) = p$ . We can then construct a disobedient equilibrium in which the first group is the last disobedient group ( $\bar{n} = 1$ ), the fraction of attack from the first group at the survival threshold is  $x^*$ , and

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<sup>27</sup>For any  $\theta' < \theta''$ , agents observe higher private signals under  $\theta''$ . Moreover, they will observe weakly higher passes under  $\theta''$ . Therefore, under a monotone equilibrium, the aggregate attack under  $\theta''$  would be lower,

$$w^{\frac{n}{N}}(\theta'') = \frac{1}{N} \sum_{n'=1}^n F\left(\frac{\hat{s}_{n'} - \theta''}{\sigma}\right) \leq \frac{1}{N} \sum_{n'=1}^n F\left(\frac{\hat{s}_{n'} - \theta'}{\sigma}\right) = w^{\frac{n}{N}}(\theta').$$

Therefore, the residual fundamentals,

$$\theta^{\frac{n}{N}}(\theta'') - \theta^{\frac{n}{N}}(\theta') = \theta'' - w^{\frac{n}{N}}(\theta'') - (\theta' - w^{\frac{n}{N}}(\theta')) \geq \theta'' - \theta'$$

the marginal agent  $\hat{s}_1$  in group 1 is indifferent:  $G(x^*, N, k) = p$ . Formally, define  $\hat{s}_1$  such that  $F\left(\frac{\hat{s}_1 - \hat{\theta}}{\sigma}\right) = x^*$  where  $\hat{\theta} = \theta_1 = \frac{x^*}{N} + k\left(1 - \frac{1}{N}\right)$ . Therefore,  $\mathcal{E}^D(\Gamma) \neq \emptyset$ . This makes **Condition C** a necessary condition for credibility.

**Step 4:** Suppose that **Condition C** is satisfied. This means for all  $x \in (k, 1]$ ,  $G(x, N, k) > p$ . Therefore, if at the survival threshold of fundamental  $\hat{\theta}$ ,  $x$  fraction of agents attack from the last disobedient group  $\bar{n}$ , then the marginal agent in the last disobedient group strictly prefers not attacking. This rules out any disobedient equilibrium; i.e.,  $\mathcal{E}^D(\Gamma) = \emptyset$ . This makes **Condition C**, a sufficient condition for credibility. ■

The following lemma is useful for proving **Theorem 2**, **Proposition 3**, and **Proposition 4**.

**Lemma A.1.**  $y(N, k)$  is strictly increasing in each of  $N, k$ , and is quasiconcave.

*Proof of Lemma A.1.*

$$y(N, k) \equiv \min_{x \in [0, 1]} \{ \tilde{G}(x, N, k) := G(x, N, k) \cdot \mathbb{1}(x > k) + \mathbb{1}(x \leq k) \} \quad (\text{A.10})$$

$\tilde{G}(x, N, k)$  is continuous. Hence,  $y(N, k)$  is well-defined and the minimizer  $x \in (k, 1)$  because for all  $x \in (k, 1)$ ,  $\tilde{G}(x, N, k) < 1$  and for all  $x \in [0, k] \cup \{1\}$ ,  $\tilde{G}(x, N, k) = 1$ . It follows that  $y(N, k)$  is increasing in  $N$  and  $k$  because  $G(x, N, k)$  is increasing in  $N$  and  $k$ . For quasiconcavity, first, we show that  $G(x, N, k)$  is quasiconcave, which makes  $\tilde{G}(x, N, k)$  quasiconcave. Therefore,  $y(N, k)$  is quasiconcave (see [Boyd, Boyd and Vandenberghe, 2004](#)). The formal proof is relegated to the Online Appendix B. ■

*Proof of Proposition 3.* From **Lemma A.1**, we know  $y(N, k)$  is increasing in  $N$ , and note that  $\lim_{N \rightarrow \infty} y(N, k) \rightarrow 1$ . Therefore, the results follows. ■

*Proof of Theorem 2.* Consider  $y(N, k = 0)$  and note that  $\lim_{N \rightarrow \infty} y(N, 0) = 1 > p$ . Since  $y(N, 0)$  is increasing in  $N$ , there exists a unique  $N_0 \geq 1$  such that  $y(n, k = 0) > p$  if and only if  $n \geq N_0$ .<sup>28</sup> Alternatively,  $N_0 = \min\{N \in \mathbb{Z}^+ | y(N, k = 0) > p\}$ . For any  $N \leq N_0 - 1$ , let  $N'$  be any number in the range of  $(N, N_0)$ . By the continuity of  $y(N, k)$  in  $k$ , we have  $y(N, \underline{k}(N)) = p = y(N', \underline{k}(N')) < y(N', \underline{k}(N))$ . By **Lemma A.1**,  $y(N, k)$  is increasing in  $k$ , it must be that  $\underline{k}(N') < \underline{k}(N)$ . Therefore, for  $N \in [1, N_0 - 1]$ ,  $\underline{k}(N)$  is strictly decreasing in  $N$ ; and, for all  $N \geq N_0$ ,  $\underline{k}(N) = 0$ . This completes the first two parts of the Proposition. Next, we prove the discrete convexity of  $\underline{k}(N)$  for  $N \in [1, N_0 - 1]$ .

<sup>28</sup>For the details of the existence of  $N_0$ , see [Basak and Zhou \(2020\)](#). For example, if  $\sigma$  is sufficiently large, then  $N_0 = 1$ .

By definition,  $\Gamma^C = \{(N, k) : y(N, k) > p\}$ . Consider  $(N, \underline{k}(N)^+), (N', \underline{k}(N')^+) \in \Gamma^C$ . By [Lemma A.1](#),  $y(N, k)$  is quasiconcave, which implies that, for  $\lambda \in [0, 1]$ ,

$$y(\lambda N + (1 - \lambda)N', \lambda \underline{k}(N)^+ + (1 - \lambda)\underline{k}(N')^+) > p. \quad (\text{A.11})$$

That means,  $(\lambda N + (1 - \lambda)N', \lambda \underline{k}(N)^+ + (1 - \lambda)\underline{k}(N')^+) \in \Gamma^C$ , proving that the set  $\Gamma^C$  is convex. By the definition of  $\underline{k}(N)$  (see (8)),  $\underline{k}(N) = \inf\{k : y(N, k) > p\}$ . Given that (A.11) implies  $\underline{k}(\lambda N + (1 - \lambda)N') \leq \lambda \underline{k}(N) + (1 - \lambda)\underline{k}(N')$ , thereby proving the convexity of  $\underline{k}(N)$ .<sup>29</sup> As  $\underline{k}$  is defined on  $\mathbb{Z}^+$ , it is discrete convex. This is because, we have shown that the extension of  $\underline{k}$  defined on  $\mathbb{R}^+$  is convex (see [Boyd, Boyd and Vandenberghe \(2004\)](#)). By the definition of discrete convexity,  $\underline{k}(N) - \underline{k}(N + 1)$  is decreasing in  $N$  for  $N \in [1, N_0 - 1]$ . ■

*Proof of Proposition 4.* The formal proof is relegated to the Online Appendix B. Below, we provide a sketch of the main argument behind the proof. Following the definition (A.10) in [Lemma A.1](#), define  $y(\sigma, N, k) = \min_{x \in [0, 1]} \tilde{G}(x, \sigma, N, k)$ . From the arguments in the proof of [Lemma A.1](#), the minimizer,  $\underline{x}(\sigma, N, k) \in (k, 1)$ . Therefore,  $y(\sigma, N, k) = G(\underline{x}(\sigma, N, k), N, k)$ . Because,  $G(x, \sigma, N, k)$  is increasing in  $\sigma$ , so is  $y(\sigma, N, k)$ . To see this, consider any  $\sigma > \sigma'$ ,

$$y(\sigma', N, k) = G(\underline{x}(\sigma', N, k), \sigma', N, k) \leq G(\underline{x}(\sigma, N, k), \sigma', N, k) < G(\underline{x}(\sigma, N, k), \sigma, N, k) = y(\sigma, N, k)$$

The first inequality is from the definition of  $\underline{x}(\sigma, N, k)$ , the second from the monotonicity of  $G(x, \sigma, N, k)$  in  $\sigma$ . Since  $y(\sigma, N, k)$  is increasing in  $k$ , it follows that  $\underline{k}(N, \sigma') > \underline{k}(N, \sigma)$ . An exactly similar argument applies for  $p > p'$  so that  $\underline{k}(N, p) < \underline{k}(N, p')$ . ■

*Proof of Theorem 3.* Notice that  $\Lambda(N, k)$  is decreasing and separable in  $N$  and  $k$  and it is quasiconcave. Therefore, given the convex persuasive constraint, there always exists a social optimal. Given any  $N$ , the regulator can always improve  $\Lambda$  by reducing  $k$ . However, if  $k \leq \underline{k}(N)$ , then the stress tests are no longer credible. Therefore, for any  $N$ , the social optimal stress test must have  $k = \underline{k}(N)^+$ . Therefore,  $\mathcal{P}_f^W(N, \underline{k}(N)^+) = \mathbb{P}(\theta < \underline{k}(N)^+) = \frac{\underline{k}(N) - \underline{\theta}}{\bar{\theta} - \underline{\theta}}$ . Substituting this in the regulator's objective function, we get

$$\Lambda(N, \underline{k}(N)^+) = B - \frac{\pi(B + \chi)(\underline{k}(N) - \underline{\theta})}{\bar{\theta} - \underline{\theta}} - C(N),$$

which is concave in  $N$  since both  $\underline{k}(N)$  (See [Theorem 2](#)) and  $C(N)$  are convex. Therefore, the maximizer  $N^s$  must be such that,  $\Lambda(N^s, \underline{k}(N^s)) \geq \max\{\Lambda(N^s + 1, \underline{k}(N^s + 1)), \Lambda(N^s - 1, \underline{k}(N^s - 1))\}$ .

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<sup>29</sup> $\underline{k}(N)$  is the set of limit points of the open and convex set  $\Gamma^C$ . Therefore the closure  $\Gamma^C \cup \underline{k}(N)$  is also convex.

1)))}. Defining the marginal cost as  $\Delta C(N) = C(N+1) - C(N)$ , this implies,

$$\frac{(\underline{k}(N^s) - \underline{k}(N^s + 1))}{\Delta C(N^s)} \leq \frac{\bar{\theta} - \underline{\theta}}{\pi(B + \chi)} \leq \frac{(\underline{k}(N^s - 1) - \underline{k}(N^s))}{\Delta C(N^s - 1)}.$$

Define  $H : \mathbb{Z}^+ \rightarrow \mathbb{R}$  as a function such that  $H(N) = \frac{\underline{k}(N) - \underline{k}(N+1)}{\Delta C(N)}$ . The above conditions can be simplified as  $\frac{(\bar{\theta} - \underline{\theta})}{\pi(B + \chi)} \in [H(N^s), H(N^s - 1)]$ . Since  $H(N)$  is weakly decreasing,<sup>30</sup> this can also be written as,

$$N^s = \min \left\{ N \in \mathbb{Z}^+ : \underline{k}(N) - \underline{k}(N+1) \leq \frac{\Delta C(N)(\bar{\theta} - \underline{\theta})}{\pi(B + \chi)} \right\} \quad (N^s)$$

■

*Proof of Corollary 1.* From Theorem 3,  $N^s$  satisfies:

$$\frac{\underline{k}(N^s) - \underline{k}(N^s + 1)}{C(N^s + 1) - C(N^s)} \leq \frac{(\bar{\theta} - \underline{\theta})}{\pi(B + \chi)} \leq \frac{\underline{k}(N^s - 1) - \underline{k}(N^s)}{C(N^s) - C(N^s - 1)}.$$

For any  $\pi' > \pi$ , we have,  $\frac{\bar{\theta} - \underline{\theta}}{\pi(B + \chi)} > \frac{\bar{\theta} - \underline{\theta}}{\pi'(B + \chi)}$ . Accordingly, either of the two cases holds true,

$$\frac{\underline{k}(N^s) - \underline{k}(N^s + 1)}{C(N^s + 1) - C(N^s)} \leq \frac{\bar{\theta} - \underline{\theta}}{\pi'(B + \chi)}, \text{ or } \frac{\underline{k}(N^s) - \underline{k}(N^s + 1)}{C(N^s + 1) - C(N^s)} > \frac{\bar{\theta} - \underline{\theta}}{\pi'(B + \chi)}.$$

In the first case,  $N^s(\pi') = N^s(\pi)$  while in the second, due to the weakly decreasing nature of  $H(N) = \frac{\underline{k}(N) - \underline{k}(N+1)}{C(N+1) - C(N)}$ ,  $N^s(\pi') > N^s(\pi)$ . This implies that  $N^s(\pi') \geq N^s(\pi)$ . An exactly similar arguments apply for  $\chi' > \chi$ , so that  $\frac{\bar{\theta} - \underline{\theta}}{\pi(B + \chi)} > \frac{\bar{\theta} - \underline{\theta}}{\pi(B + \chi')} \Rightarrow N^s(\chi') \geq N^s(\chi)$ . ■

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<sup>30</sup>To see this, consider  $(\underline{k}(N-1) - \underline{k}(N)) - (\underline{k}(N) - \underline{k}(N+1)) = 2 \left( \frac{\underline{k}(N+1) + \underline{k}(N-1)}{2} - \underline{k}(N) \right) \geq 0$ . Since  $\underline{k}(N)$  is convex,  $\alpha \underline{k}(N+1) + (1-\alpha) \underline{k}(N-1) \geq \underline{k}(\alpha(N+1) + (1-\alpha)(N-1))$  for all  $\alpha \in [0, 1]$  and particularly for  $\alpha = 0.5$ . Combining this with the convexity of  $C(N)$ , i.e.  $\Delta C(N) \geq \Delta C(N-1)$ ,  $H(N) - H(N-1) \leq 0$ .